



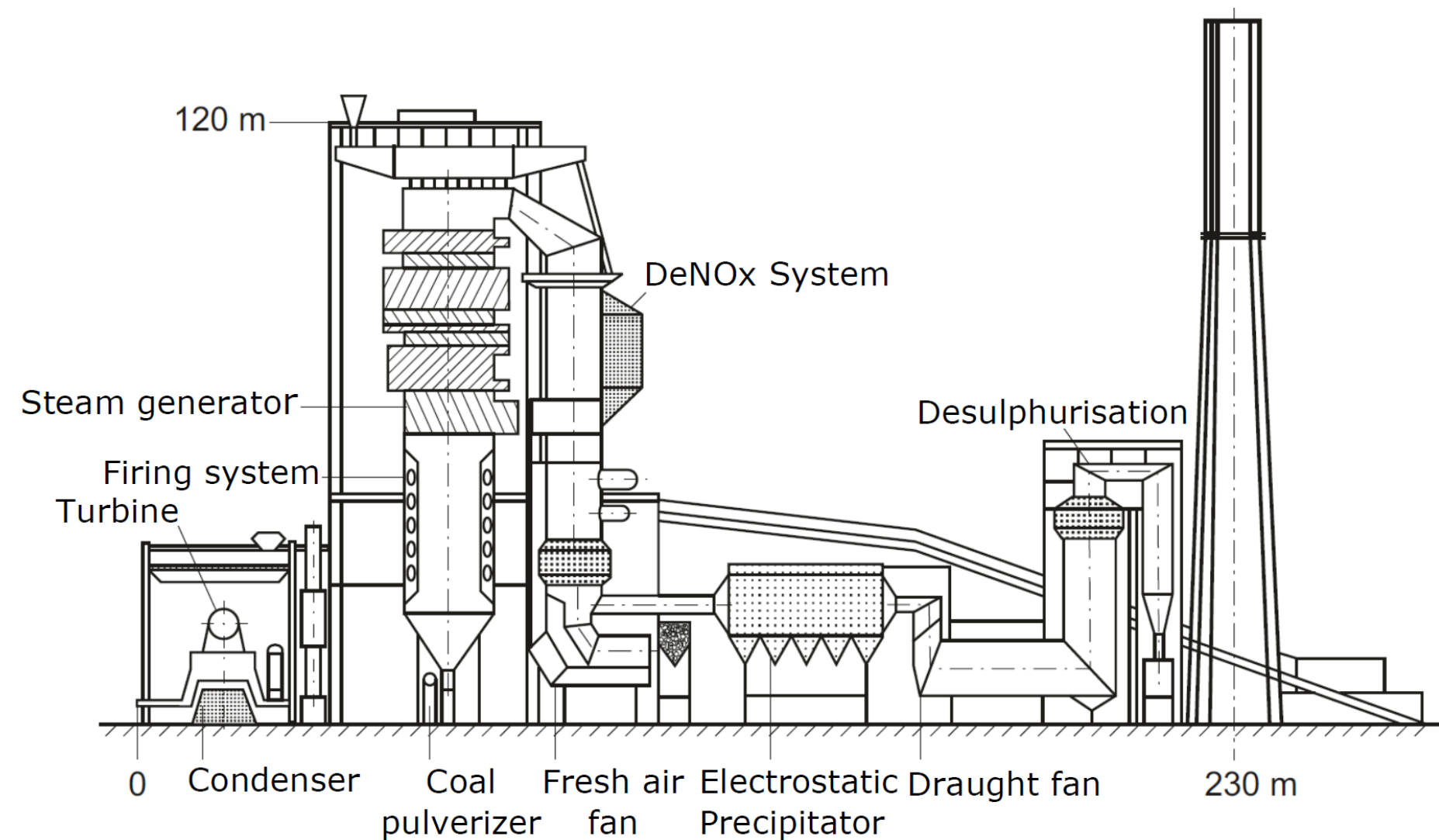
Andreas Nenning
Winter Term 2026
Course Number 074742056201

Thermal Power Plants



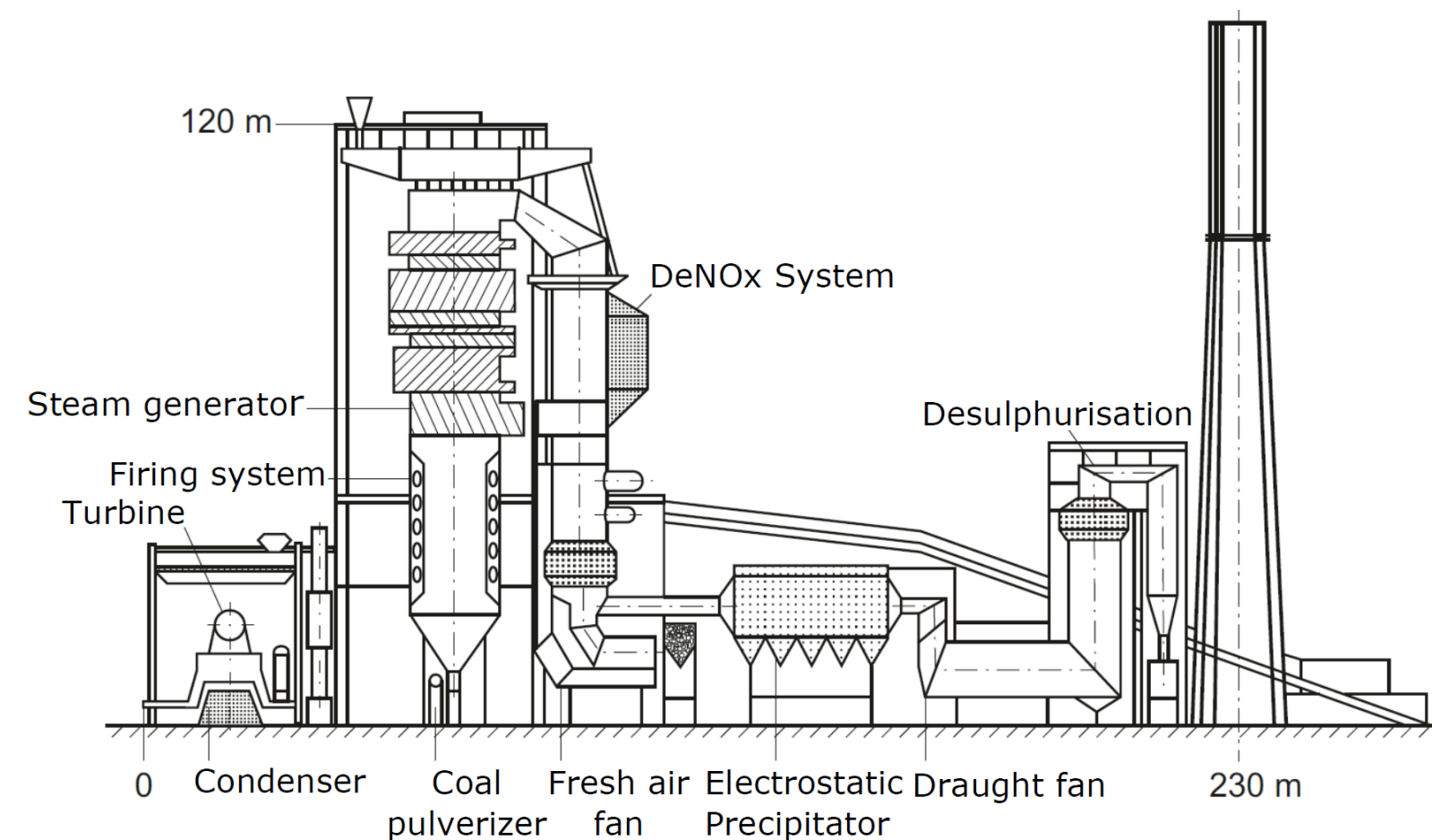
Coal Fired Power Plants

- ◆ Operates on the Rankine cycle
 - including super heating
 - modified feed-water heating
- ◆ High efficiency by using highest possible temperature and pressure
- ◆ Thermal efficiency of 40%



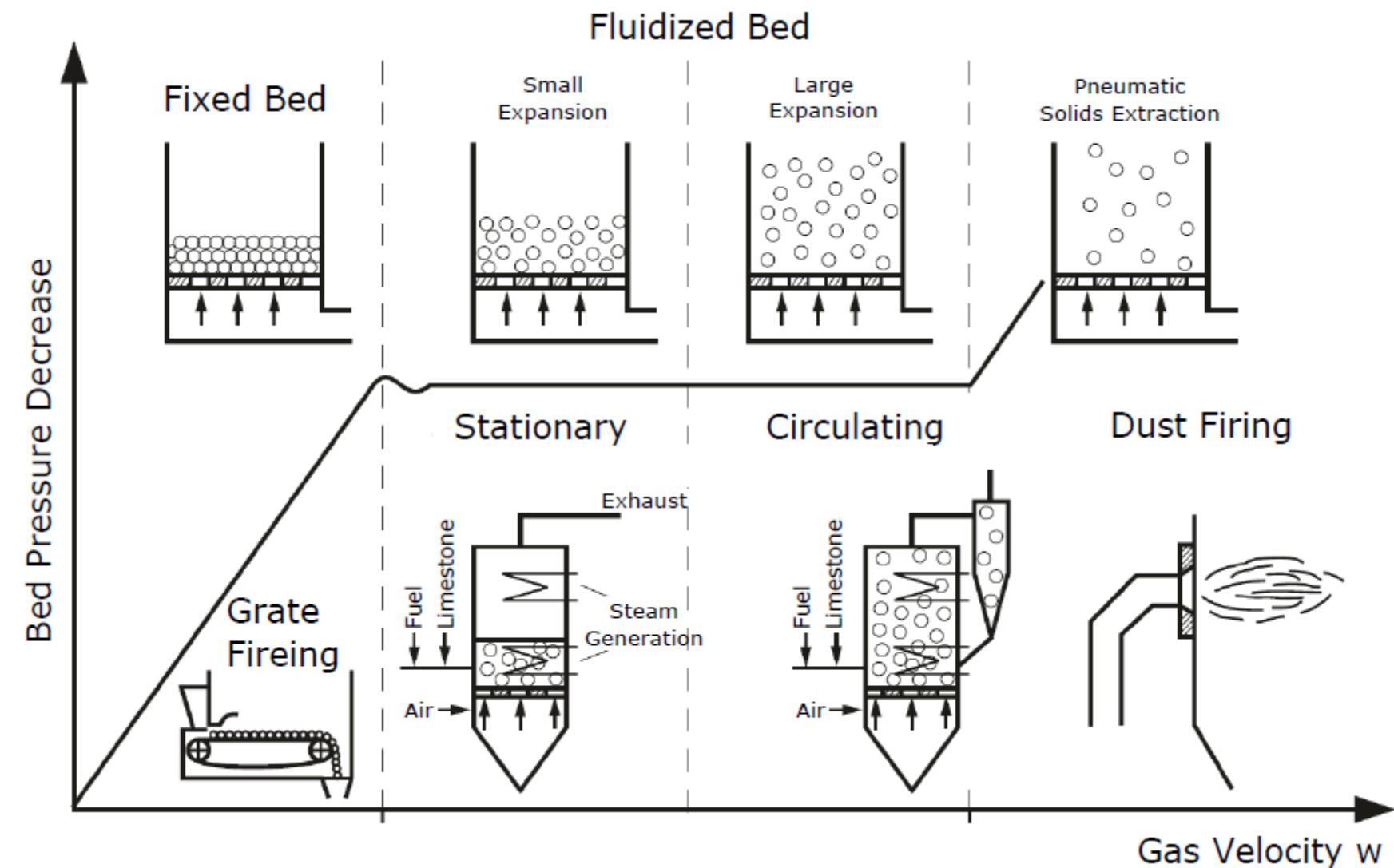
Coal Fired Power Plants

- ◆ Firing system (dt. Feuerung)
- ◆ Coal pulverizer (dt. Kohlemühlen)
- ◆ Combustion air fan (dt. Frischluftgebläse)
- ◆ Steam generator (dt. Dampferzeuger)
- ◆ Turbine
- ◆ Condenser (dt. Kondensator / Verflüssiger)
- ◆ DeNOx system (dt. DeNOx- Anlage)
- ◆ Electrostatic Precipitator (dt. Elektrofilter)
- ◆ Draught fan (dt. Saugzuggebläse)
- ◆ Desulphurisation (dt. Entschwefelung)



Firing System

- ◆ Fossil fuels are mixed with air and burned
- ◆ Categorization based on degree of suspension
 - Fixed bed or grate firing (dt. Rostfeuerung)
 - Fluidized-bed firing system (dt. Wirbelschichtfeuerung)
 - Dust firing (dt. Staubfeuerung)

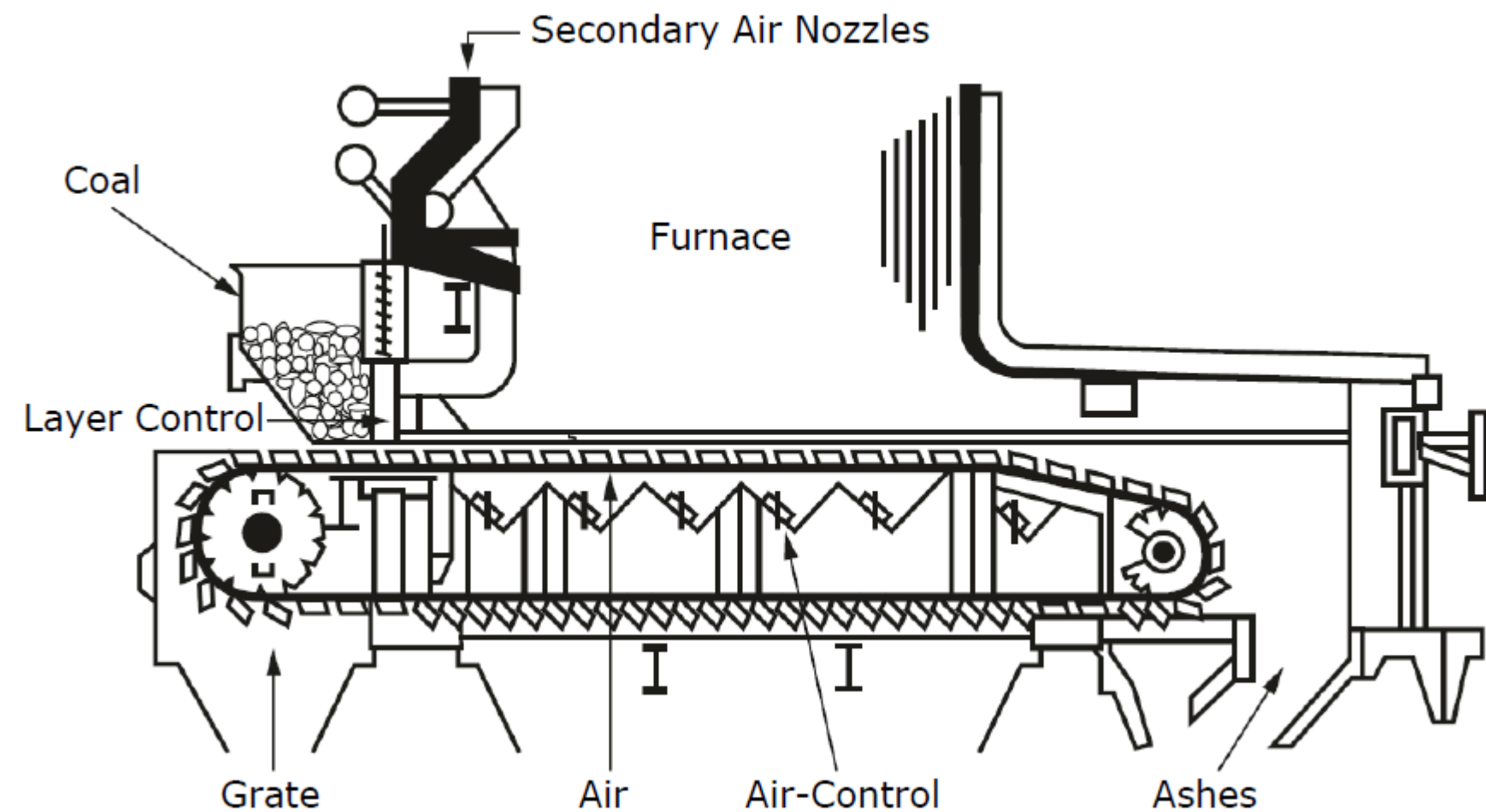


Grate firing system

- ♦ Grates widely used prior commercial use of fluidized bed
- ♦ Possible to burn a wide range of fuels
- ♦ Still in use, but implemented as moving grate
- ♦ Conveyor like setup move the fuel through the combustion chamber

Grate firing system

- ♦ Firing process consists of following steps
 - Feeding into the moving grate
 - Drying
 - Ignition by the heat within the vessel
 - Combustion
 - Removing of the ashes

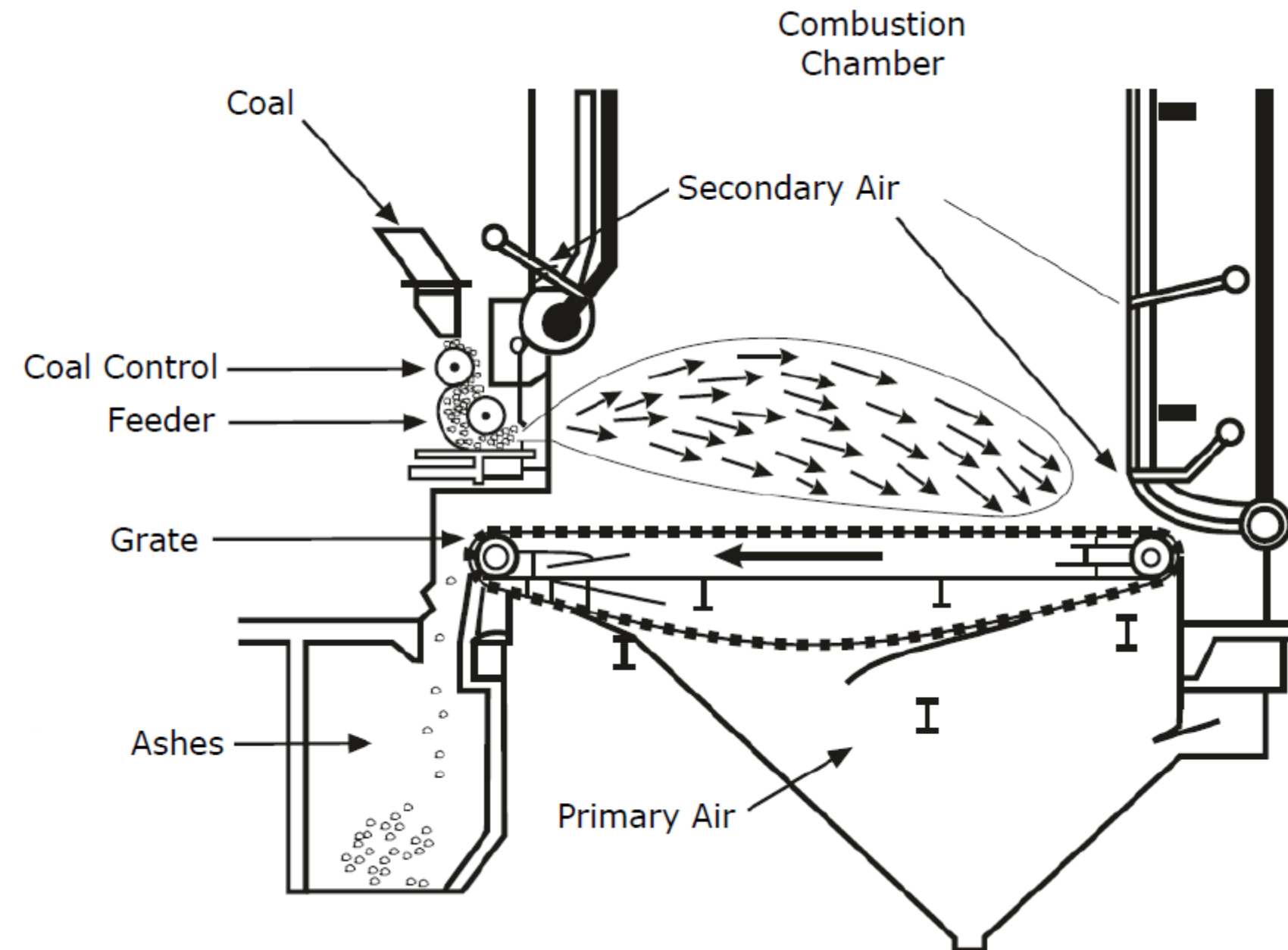


Grate firing

- ◆ Limitations
 - Controlling output power
 - Processing small particle sizes
- ◆ Power control
 - done by changing the speed of the moving grate
 - Control range +/- 10% per minute
- ◆ Fuel particles less than 1 mm are falling through the grate
 - increasing the unburned fuel losses

Feed Stoker (dt. Wanderbeschickung)

- ◆ Feed Stoker overcome grate firing limitations
- ◆ Fuel particles are thrown onto the inverse moving grate
- ◆ Most of the fuel already burns before reaching the grate
- ◆ Cooling of the grate can significantly be reduced
- ◆ Fuel is ignited by the burning material on the grate
- ◆ Higher and faster load changes can be realized



Feed Stoker

- ♦ Very suitable for fuel with high percentage of water
 - raw lignite
 - peat (dt. Torf)
 - wood
 - waste incineration plant (dt. Müllverbrennungsanlage)

- ♦ Grate firing in general can be used up to 100 MW power output
 - if a feed stoker is used: up to 150 MW

Feed Stoker

♦ Advantages are

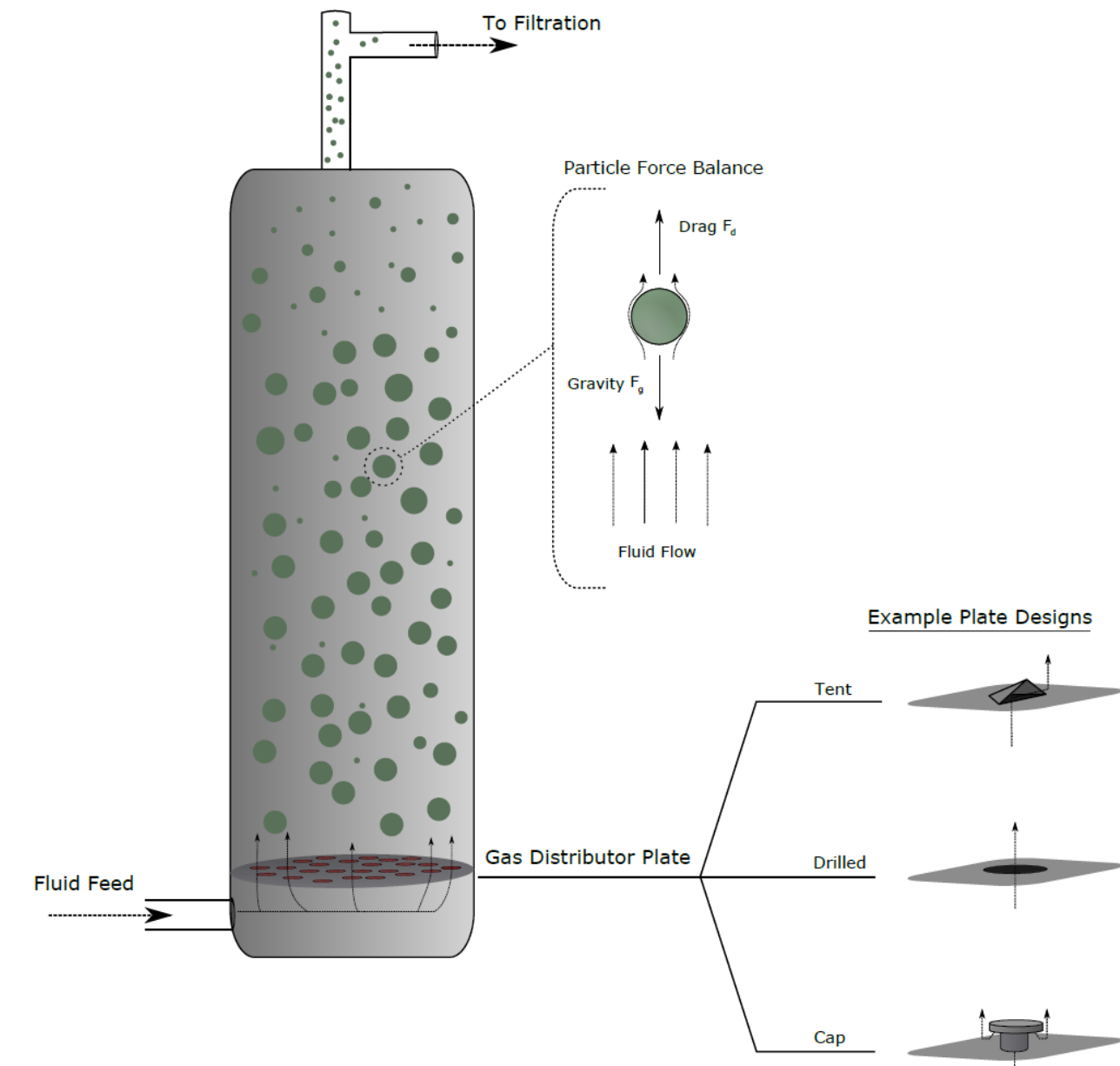
- Clear structure and high availability
- Simple operation and low maintenance costs
- Positive partial load and load change behavior
- Possibility to bind sulfur (dt. Schwefel) by adding limestone (dt. Kalk) into the combustion chamber (up to 70%)
- Relatively low NO_x (dt. Stickoxid) emission
- Low auxiliary power supply needed

♦ Disadvantages

- High unburned fuel losses (2-4%)
- Air pre-heating limited and thus high exhaust temperature ($\sim 150^\circ\text{C}$)
- Not suitable for fine coal

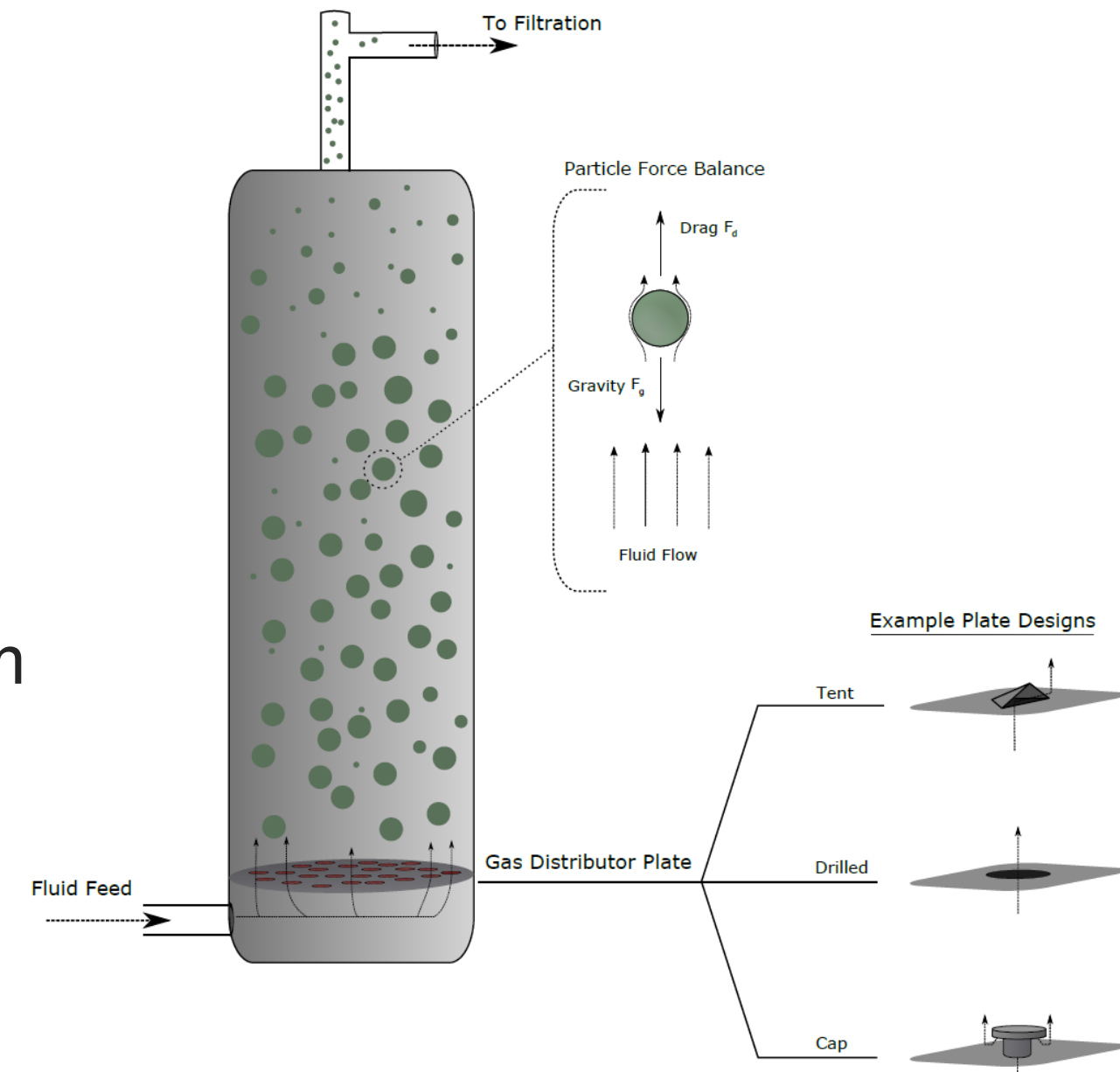
Fluidized Bed Combustion (dt. Wirbelschicht Verbrennung)

- ◆ Fluidized bed is a
 - solid particulate substance
 - kept in a vessel
 - having pressurized gas going through it
- ◆ Results in a medium
 - similar properties like a normal fluid
 - Objects with high density will sink
 - Objects with lower density will float



Fluidized Bed Combustion

- ◆ Density change by
 - altering the gas
 - altering the solid fraction
- ◆ Advantage
 - particles have more contact surface to applied gas
 - have better thermal transport inside firing system
- ◆ Combustion bed material
 - 3 to 5% coal
 - Rest is limestone and ashes



Fluidized Bed Combustion

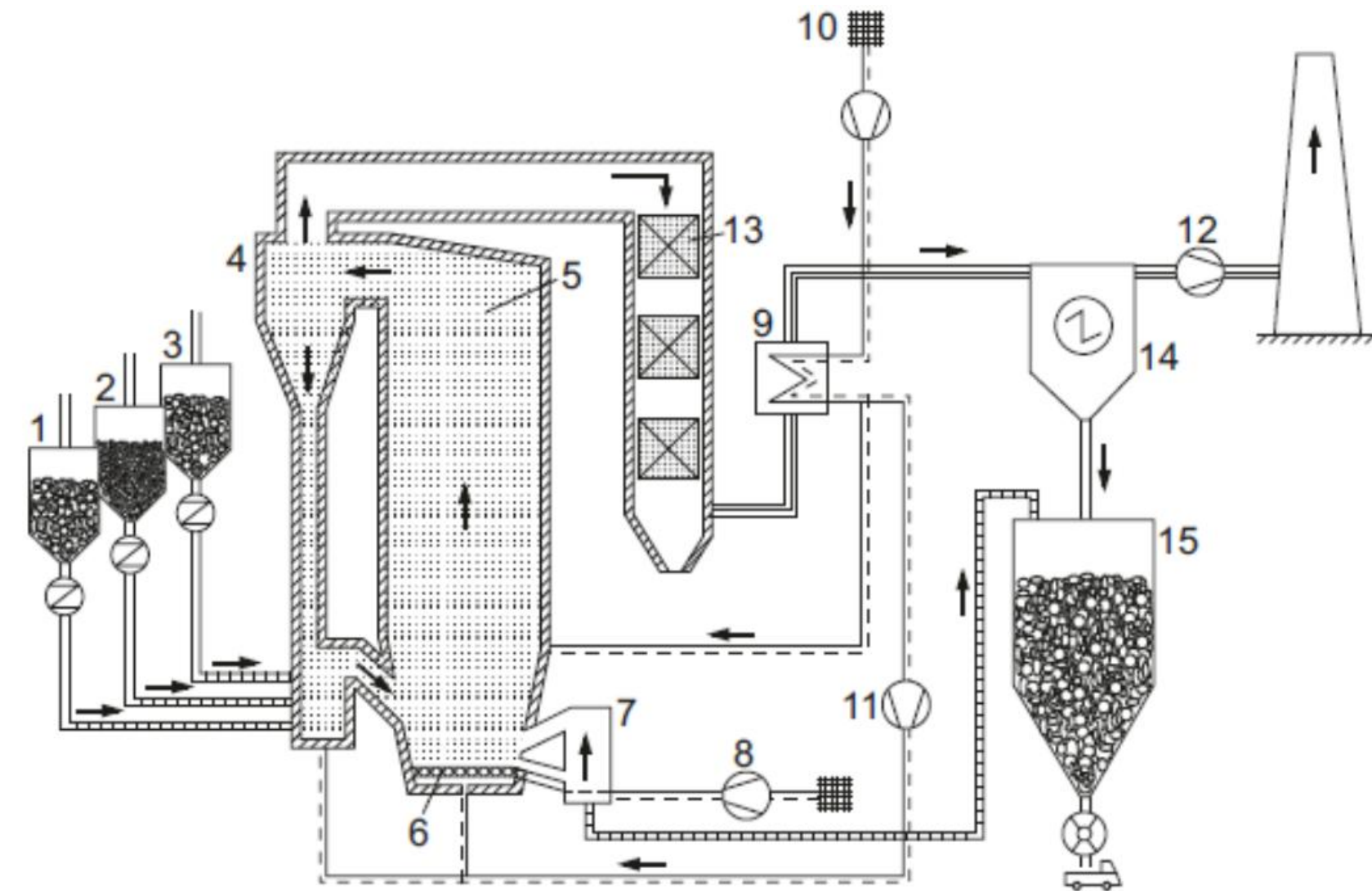
- ♦ Capable of firing a wide range of solid fuels
- ♦ Much of the ashes and sulfur is removed during the combustion
- ♦ Relatively low combustion temperature (1075 – 1225 K) results in a lower NO_x emission

Fluidized Bed Combustion

- ♦ Depending on the speed of the air flowing through the fluidized bed
 - **Solid bed** (dt. Festbett or Schüttbett)
 - Stationary or **bubbling bed** (dt. Stationäre Wirbelschicht)
 - **Circulating fluidized bed** (dt. zirkulierende Wirbelschicht)

Categorized bed types

- ♦ Bubbling fluidized bed
 - Gas at low velocities ($\sim 1\text{ m/s}$)
 - Fluidization of solid substance is stationary with some fine particles leaving the layer
- ♦ Circulating fluidized bed (CFG)
 - Gas at higher velocities ($\sim 4\text{--}8\text{ m/s}$)
 - Suspended particle bed
 - Particles are re-circulated via an external loop back into the combustion bed



1 Limestone
2 Coal
3 Ashes
4 Recirculation System
5 Combustion Chamber

6 Nozzles
7 Ash Cooler
8 Cooling Fan
9 Air Pre-Heating
10 Fresh Air

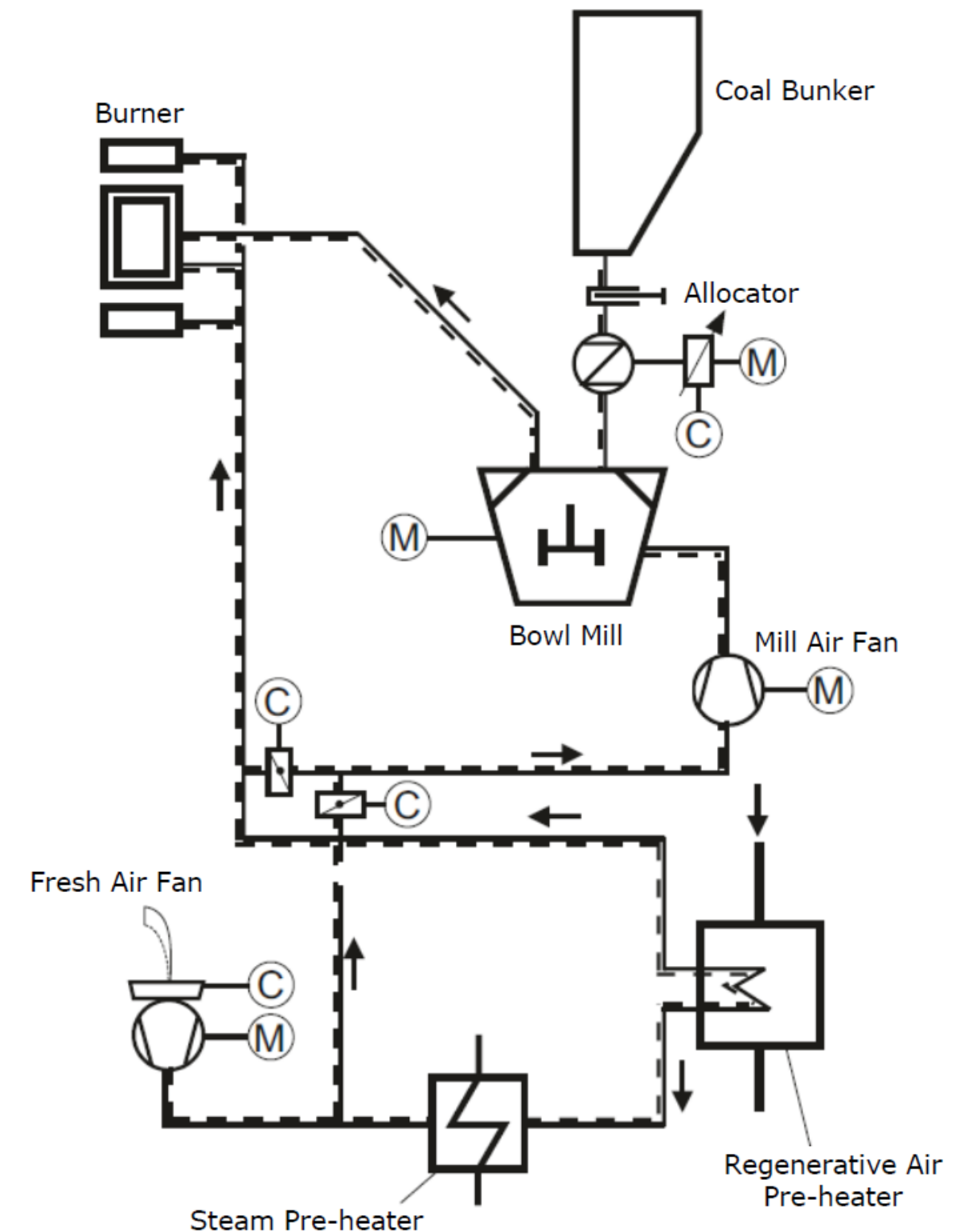
11 Primary Air Fan
12 Draught Fan
13 Super Heater
14 Dust Filter
15 Ash bunker

State of the art

- ♦ Fluidized bed technology has been introduced 1970 and is nowadays widely used
- ♦ The stationary bed types have been substituted by the circulating fluidized bed technology

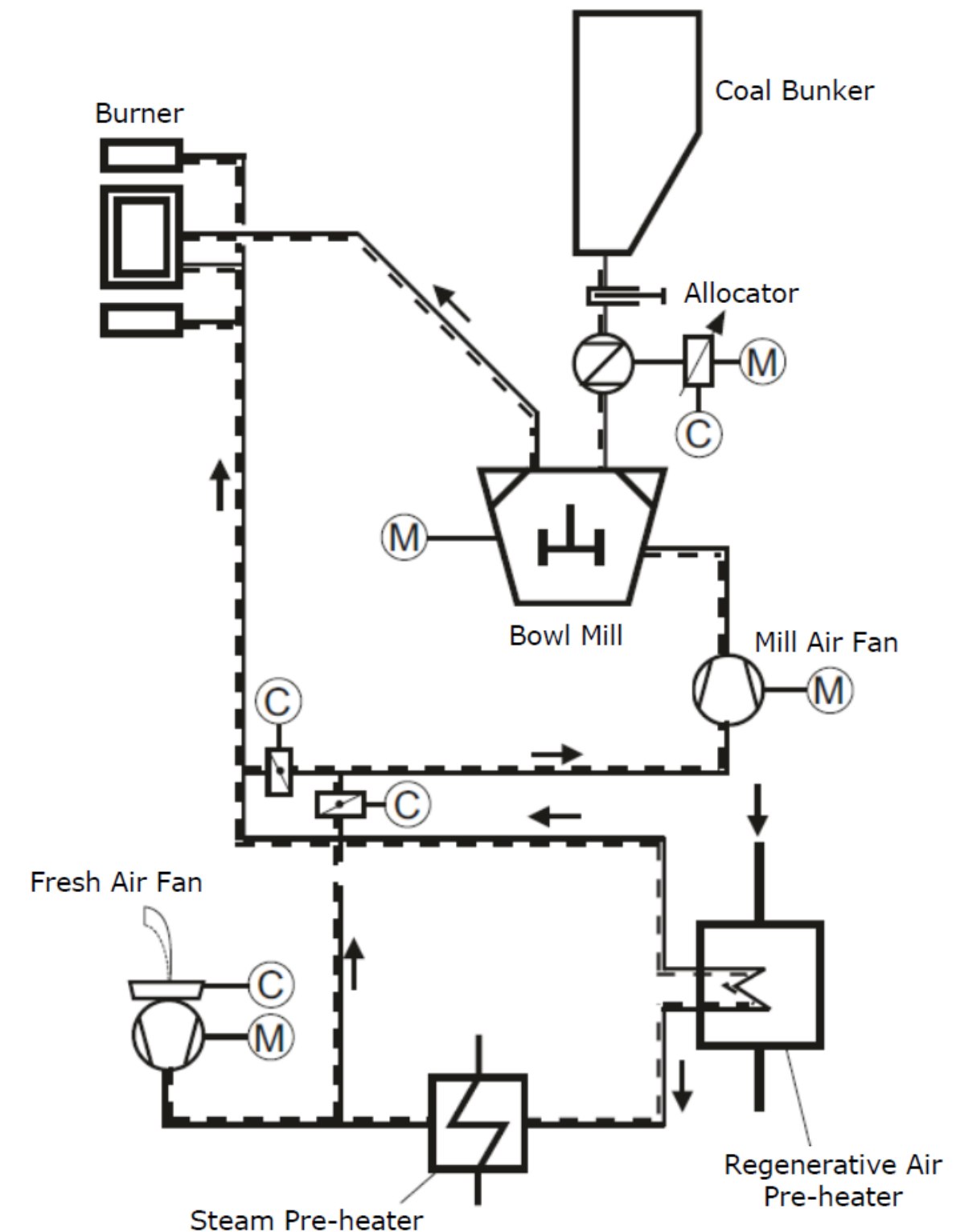
Dust Firing

- ◆ Pulverized coal has been used since 1920
- ◆ By 1950s dust firing was the main method for coal firing
- ◆ Motivation
 - treat it like oil or gas
 - transport it in pipes
- ◆ Advantages (optimized process control)
 - Fast start up
 - Can handle major load changes
 - Wide variety of fuel



Dust Firing

- ◆ Disadvantages
 - high power demand to grind the coal
 - Higher SOx and NOx emissions
 - Higher maintenance efforts (more complex)
- ◆ Raw coal grounded in coal mill
- ◆ Dried by hot air
- ◆ Blown into the combustion chamber via burner

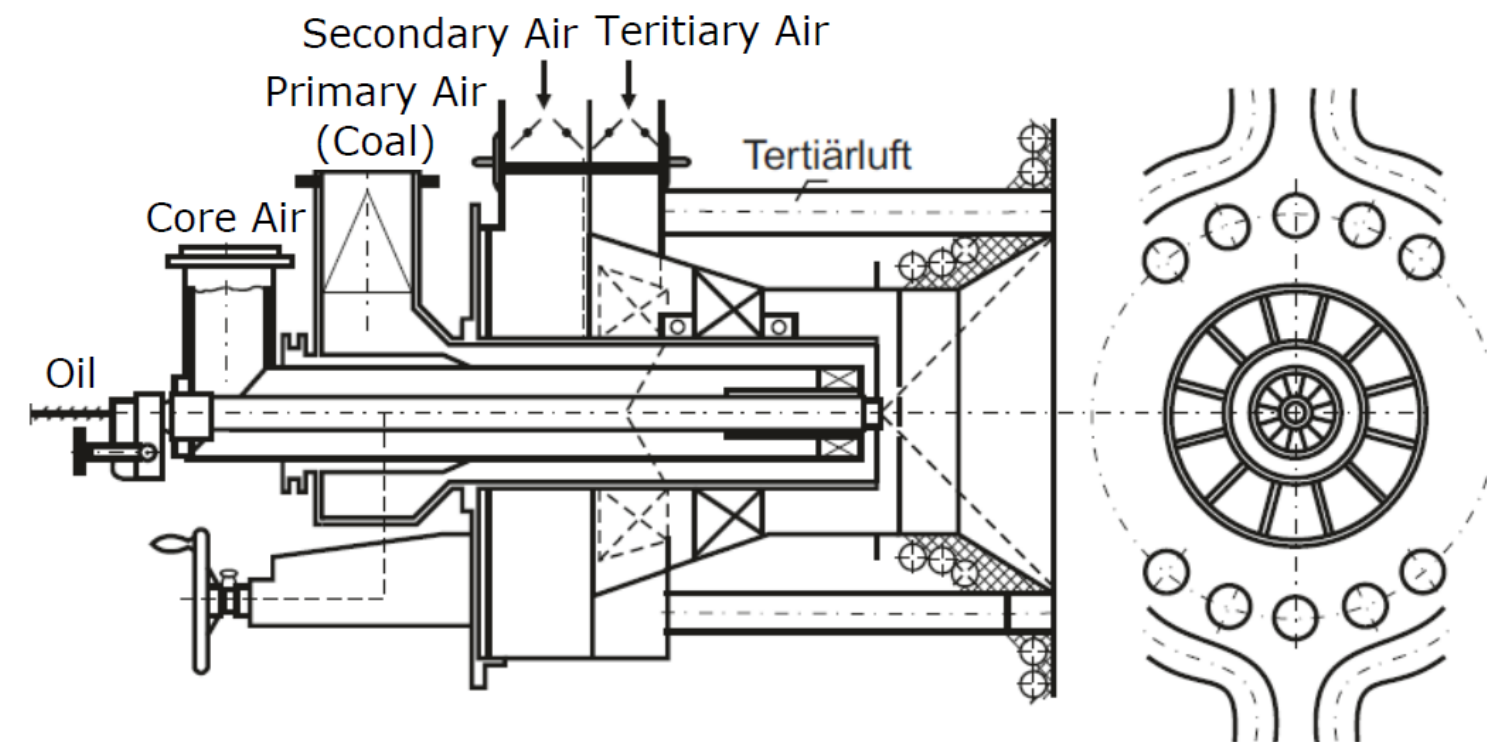


Combustion Chamber

- ♦ Combustion chamber of a 750 MW hard coal power plant
 - cuboid with 60 m height and 18 m wide
 - 32 burner are blowing more than 210 t pulverized coal per hour
 - 2 Mio m³ air per hour
 - Burning zone itself is 25 m high

Coal Burner

- ♦ Primary air (20%)
 - air which transports coal
- ♦ Secondary air (89%)
 - injected around or near the burner
- ♦ Burners can be installed
 - single side (1x)
 - opposite site (1x front – 1x rear)
 - in the corners (tangential, 4x)
- ♦ Ignition
 - with the help of an oil burner

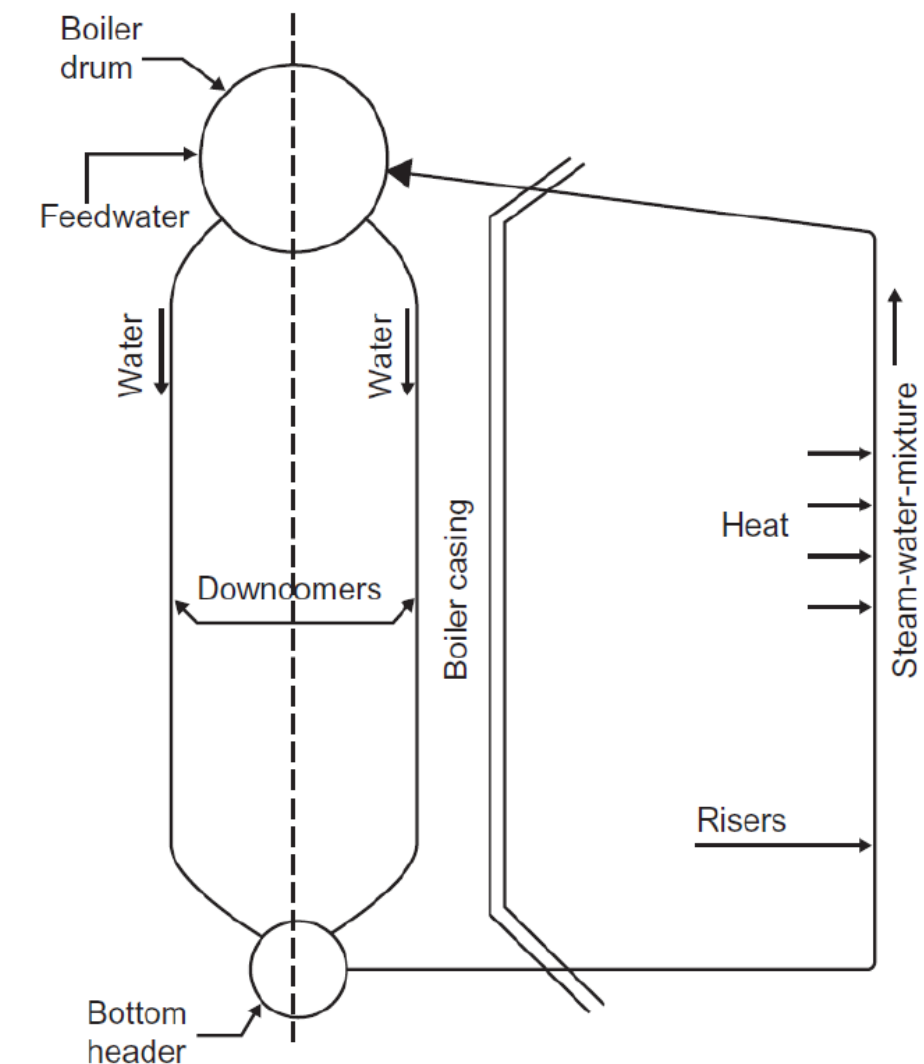


Steam generator (aka boiler)

- ♦ Steam generator is the place where the heat gets applied to the water
- ♦ Two different designs
 - Forced circulation
 - Natural circulation
- ♦ Important parameter *circulation ratio*
 - *quantity of water in circulation / quantity of steam produced*
- ♦ *Natural circulation: ratio is between 4 and 30*
- ♦ *Forced circulation: ratio is between 3 and 10 (less water needed)*

Natural Circulation

- ◆ Oldest procedure to generate steam
- ◆ Feed water flows through *downcomer* to the bottom of the furnace
- ◆ Riser tubes, heated by the furnace walls, let the steam wet steam rise towards the boiler drum
- ◆ There the steam gets separated from the water
- ◆ Steam moves further to the super heater whereas the water flows back to the bottom heater
- ◆ Because the flow is driven by the natural density differences, the method is known as the natural circulation in boilers



Forced Circulation

- ♦ Pumps are incorporated in the down flowing water to overcome friction losses
- ♦ Because the pump overcomes and friction resistance, smaller tube diameter can be used which reduces the cost of the tube material.

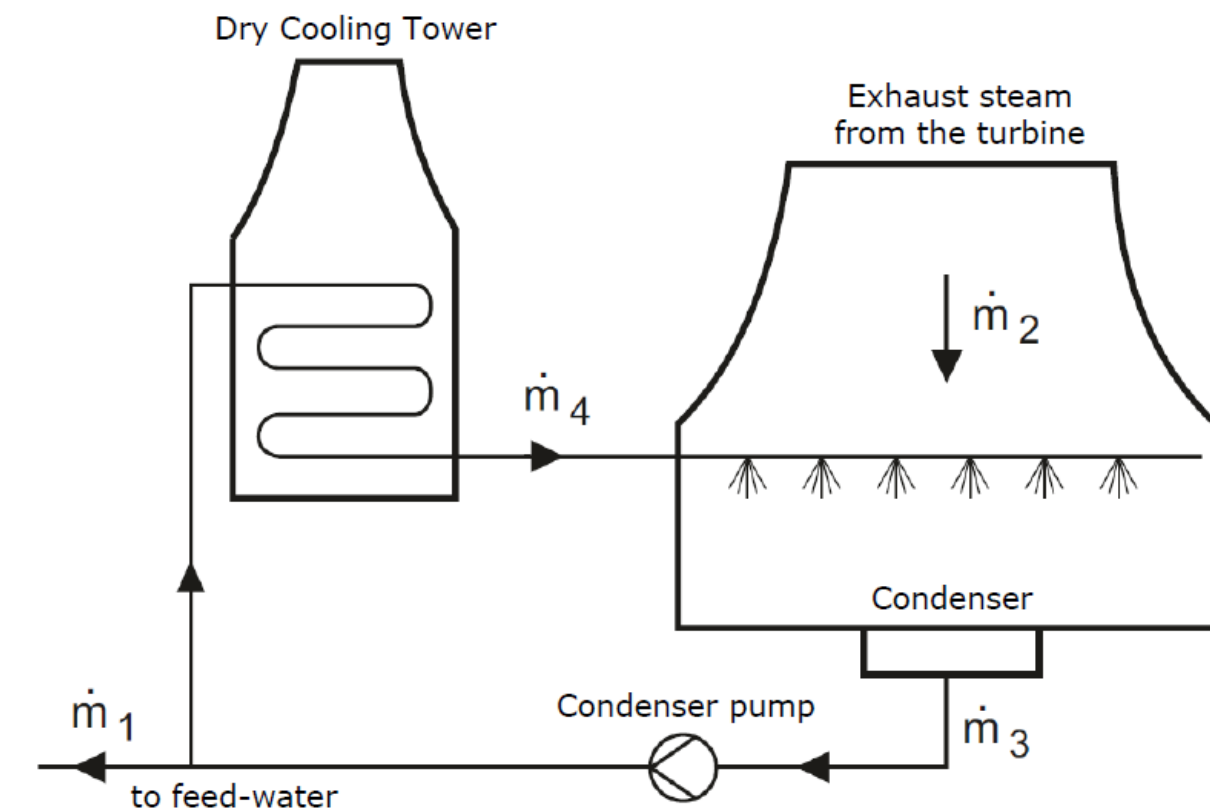
Condenser

- ◆ Condenser close the thermodynamical process
- ◆ Converting the wet steam coming from the turbine to feed water for the boiler
- ◆ Condensing is the inverse of the steam generation process
- ◆ Therefore we see a big volume change

- ◆ Condenser can be classified
 - Mixed condenser (dt. Mischkondensatoren)
 - Surface condensation (dt. Oberflächenkondensator)

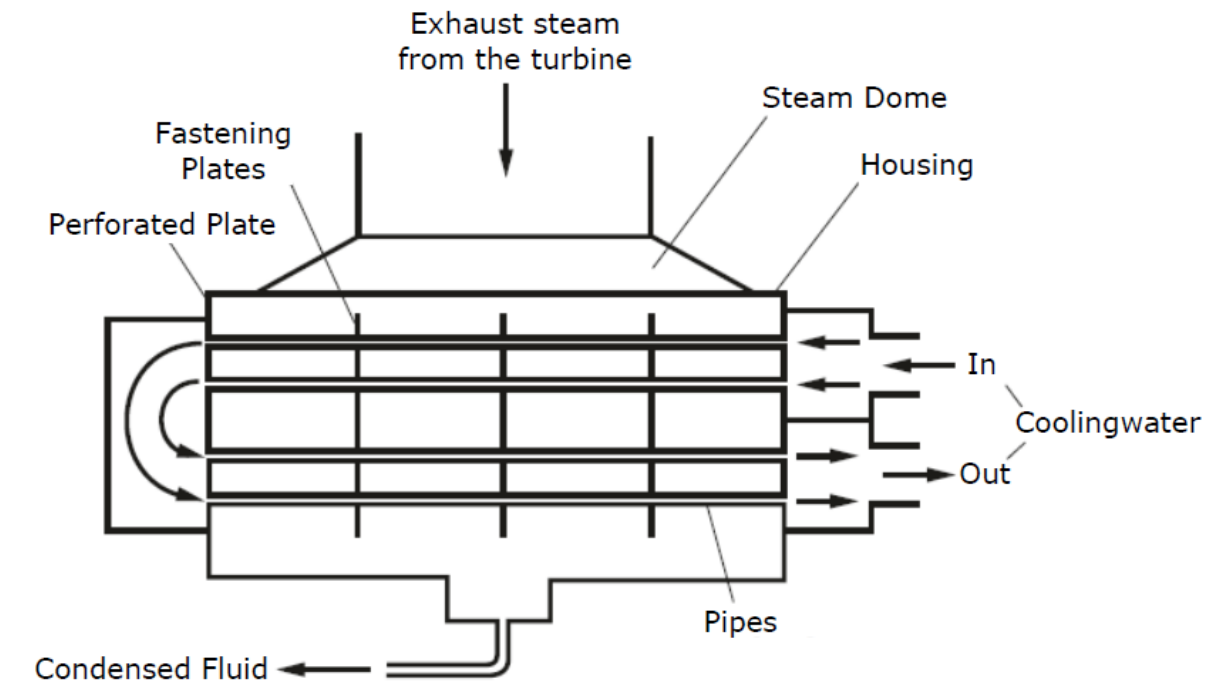
Mixing condenser

- ♦ Cooling is done by generating finest water droplets
 - Size of 0.6 mm
 - Velocity 15 m/s
- ♦ Heat transition coefficient is 100 kW/m²K



Surface condenser

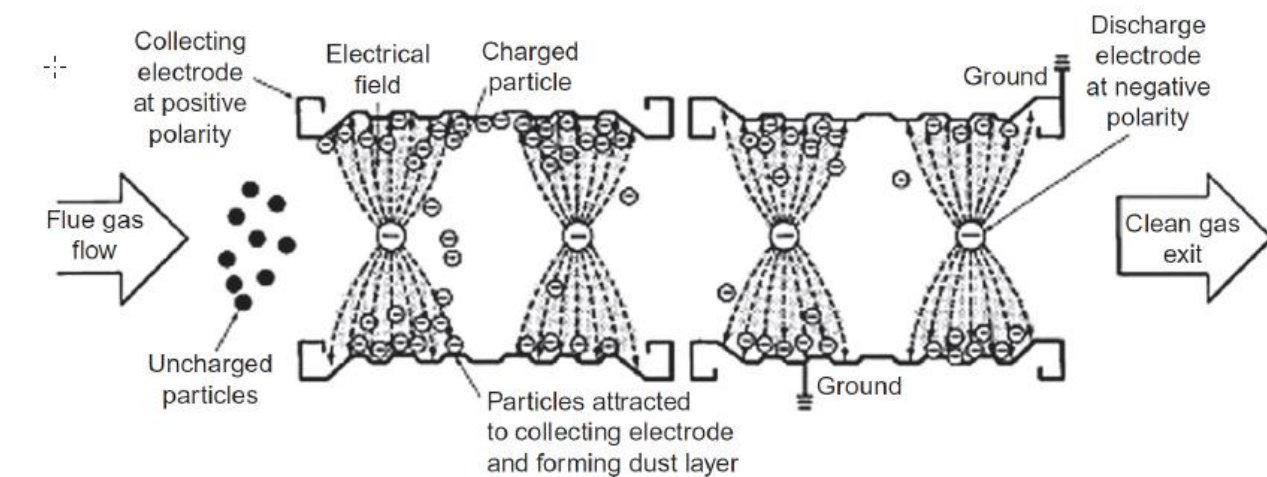
- ♦ Steam and the coolant are separated by an cooling surface
- ♦ Coolant can ether be water or air
- ♦ It is called fresh water cooling if water from a river or sea is used



Flue gas cleaning systems (dt. Rauchgasreinigungsanlage)

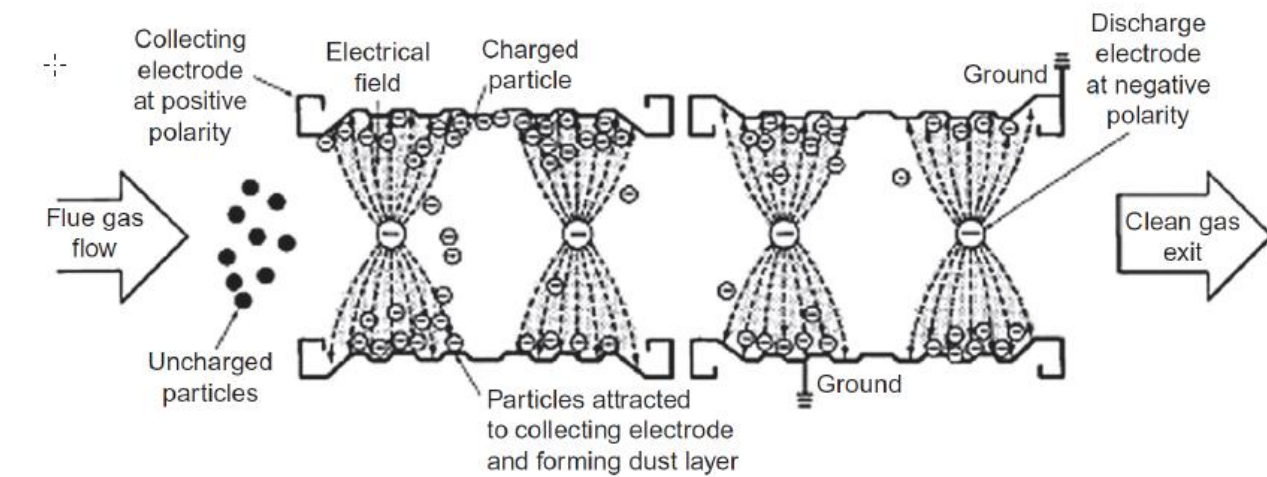
Electrostatic precipitator (ESP) (dt. Elektrostaubfilter)

- ◆ Removes small ash particles
 - 0.1 to 10 μm
- ◆ Main components
 - two sets of electrodes
- ◆ First plate (collecting electrode)
- ◆ Second plate (discharge electrodes)
- ◆ Between them a high voltage DC source forms a electric field
- ◆ Gas passes through – gets ionized – moves to collecting electrode



Electrostatic precipitator (ESP) (dt. Elektrostaubfilter)

- ◆ Whole process
 - Charging of suspended particles
 - Collection of charged particles in an electric field
 - Removal of the particles at the collecting electrodes

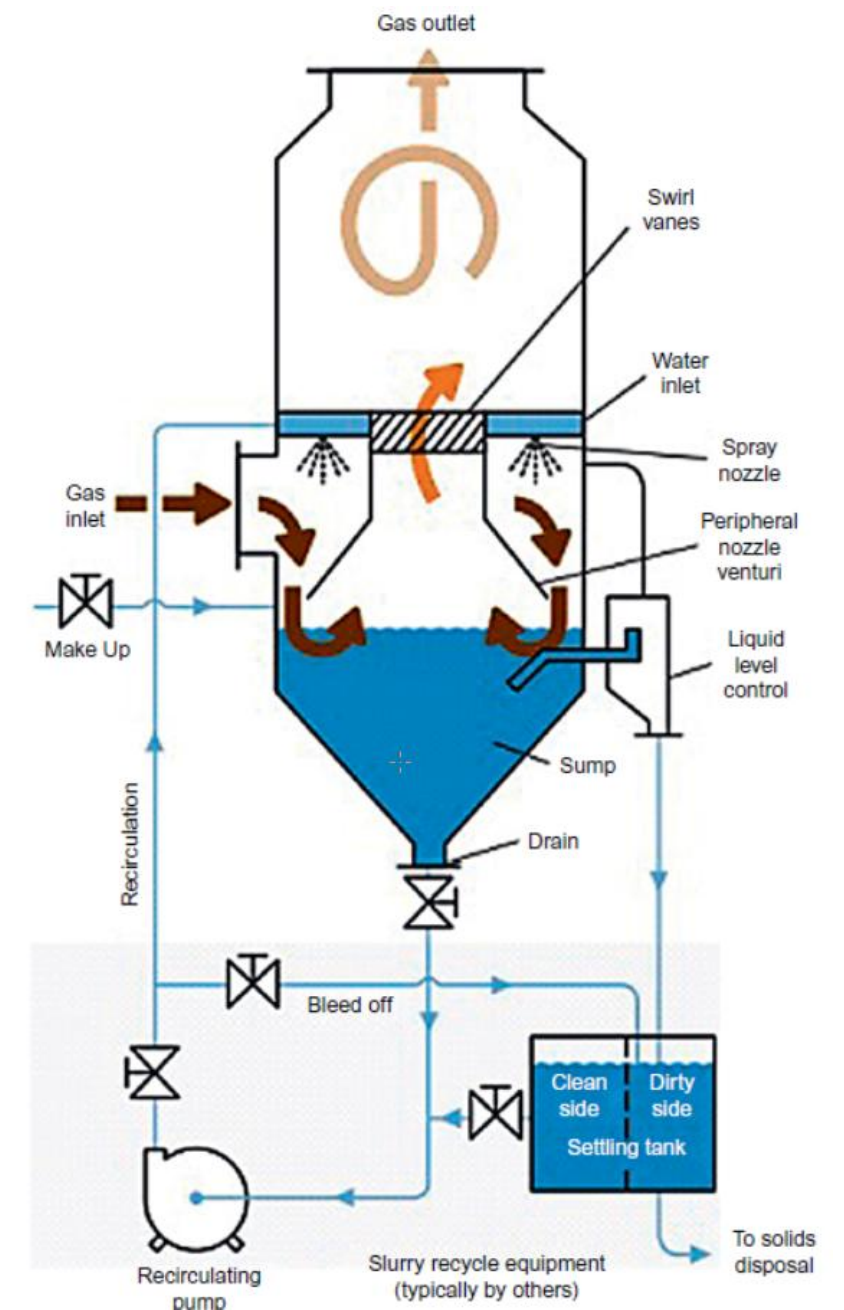


Flue-gas desulphurization (FGD) (dt. Entschwefelungsanlage)

- ♦ SO₂ gas is acidic
- ♦ Alkaline chemicals are used to remove them
- ♦ Used Alkaline chemicals
 - Limestone
 - Hydrate lime
 - Magnesium hydroxide
 - Natural alkaline like seawater
- ♦ Several different types of desulphurization technologies, *wet scrubber* is most widely used

Wet scrubber

- ♦ Flue gas enters the bottom of absorber tower
- ♦ Exits as clean gas from the top
- ♦ Absorption and oxidation take place within the tower
- ♦ Water is sprayed through spray nozzles
- ♦ Air is injected through the tower from the bottom at a lower point than the flue gas
- ♦ Mixture of calcium-, sodium-, and ammonium based sorbent are injected to the tower to react with the SO_2
- ♦ Limestone is widely used in operating wet scrubber



NO_x Emission Control (dt. Stickoxid)

- ♦ Optimizing combustion process
- ♦ Selective catalytic reduction (SCR)
- ♦ Selective non-catalytic reduction (SNCR)

Selective catalytic reduction SCR

- Ammonia vapor is used as reducing agent
- Converting NO_x into H_2O and N
- Vapor is injected into flue-gas stream if catalyst (dt. Katalytisch) is used (573 – 673° K)
- If no catalyst is used, reaction takes place between 1143 and 1478° K
- Most widely used catalyst: vanadium pentoxide in combination with titanium dioxide
- SCR can reduce NO_x up to 80% from flue gas

Selective non-catalytic reduction

- ♦ Reagent (dt. Reagenzien) is injected into the flue gas in the furnace within the appropriate temperature zone
- ♦ No catalytic used
- ♦ Reagents are
 - Ammonia
 - Urea, etc
- ♦ Process achieves 40 to 70% NO_x reduction
- ♦ Difficult to use in larger fossil fired furnaces or gas turbines
- ♦ Widely applied to fluidized bed boilers

Steam Turbine

- ◆ Machine where the steam relaxes and causes a rotary motion
- ◆ Converts potential energy to kinetic energy and then into mechanical energy
- ◆ Called flow machines (st. Strömungsmaschinen)
- ◆ Same category as
 - Wind turbines
 - Gas turbines
 - Water turbines

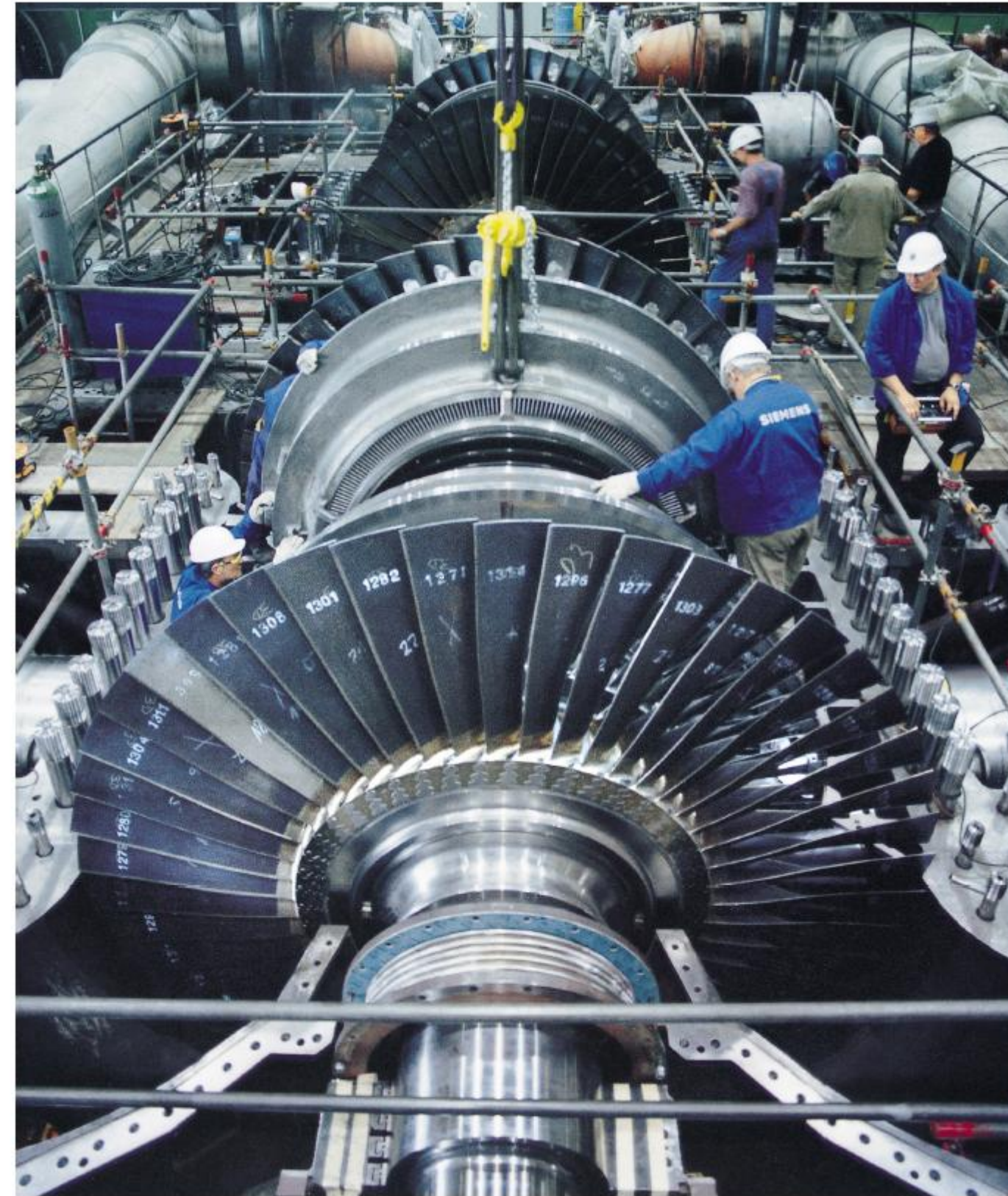


Figure 56: Assembly of a steam turbine

Components of a Turbine

- ◆ Nozzles (dt. Düsen, Leitrad)
- ◆ A rotor disc (dt. Rotor, Laufrad)
- ◆ Casing (dt. Gehäuse)

- ◆ Turbine stage
 - Nozzles
 - rotor disc
- ◆ Turbine consists of a number of stages mounted on the same rotor shaft.

Steam turbines

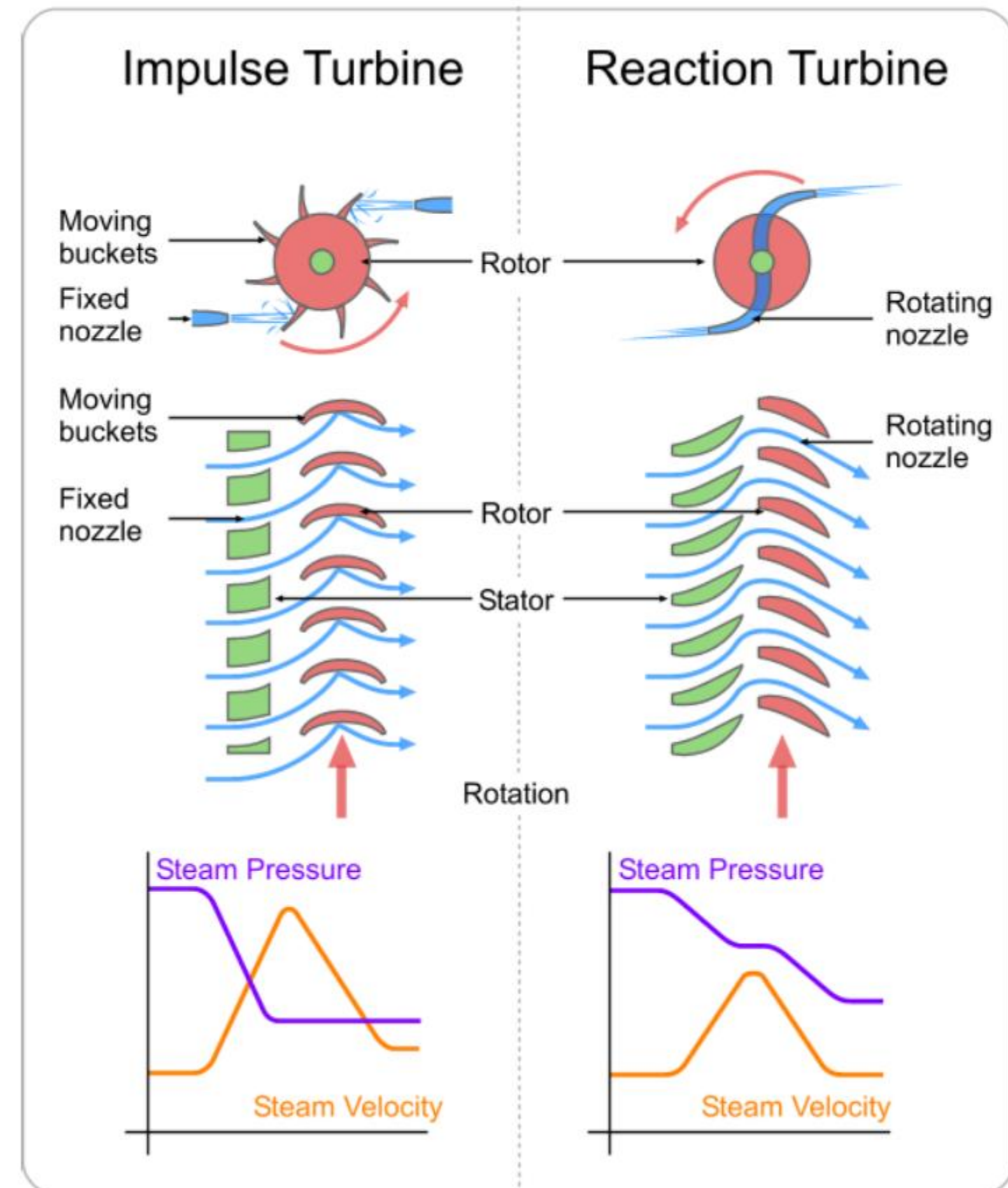
- ♦ Working medium flows in axial direction
- ♦ Nozzles are used to increase the speed of the working medium and re-direct to disc
- ♦ In reaction to the jet the disc start to spin
- ♦ Two basic working principle
 - Reaction principle
 - Impulse principle
- ♦ Difference between these two principles are how the enthalpy changes within the turbine stage

Reaction principle

- ♦ Reaction turbines (dt. Überdruckturbinen)
- ♦ Steam is expanding equally
 - in the stationary blades
 - in the rotary blades
- ♦ This type makes use of the reaction forces
 - steam accelerating through the rotor blades because of their specific shape
- ♦ Steam is directed on to the rotor by fixed blades
- ♦ Leaves stator as jet – reaches rotor blades – changes direction and increases its speed relatively to the moving blades

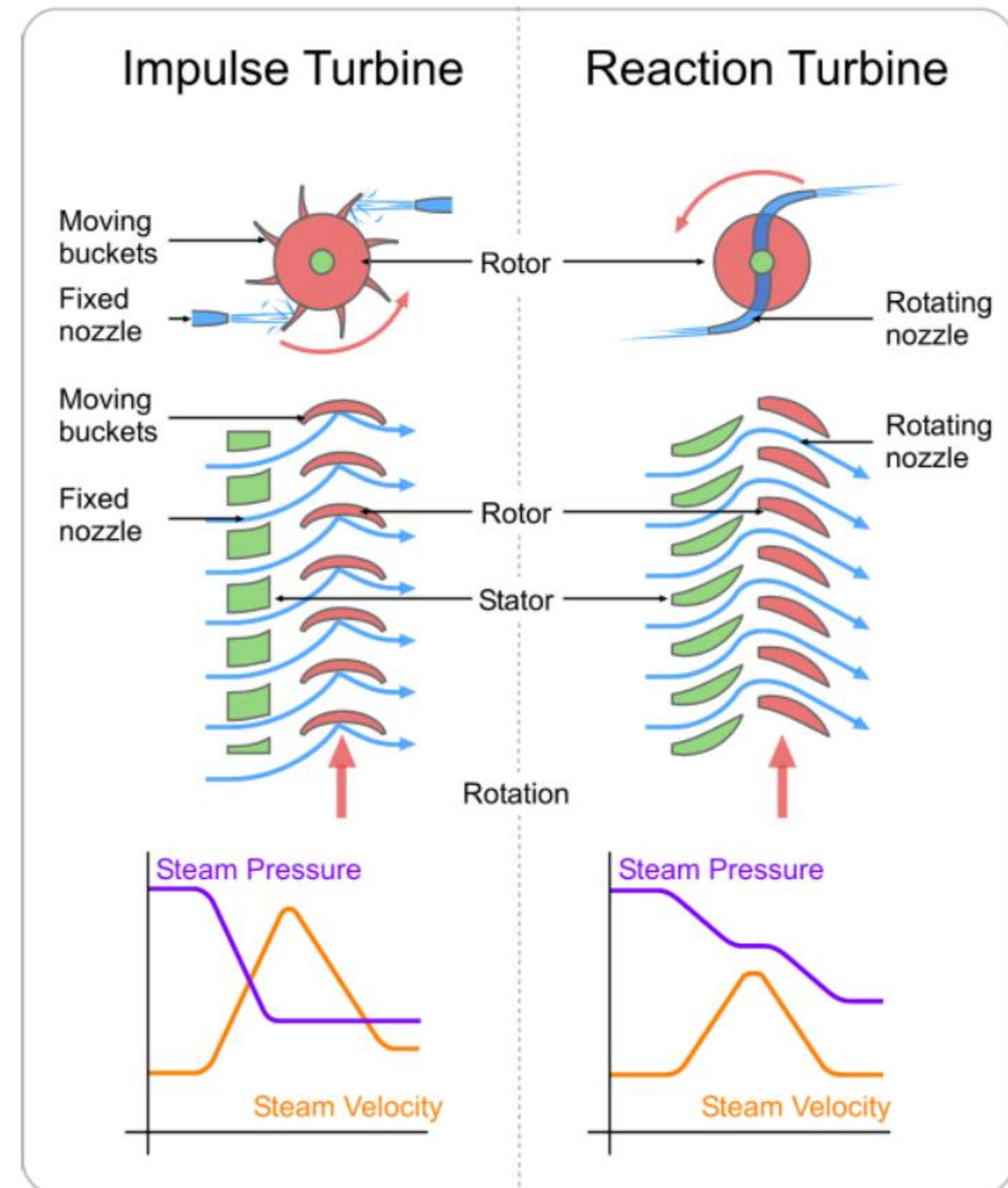
Reaction principle

- ◆ Pressure drop across stationary and moving blades
- ◆ Steam accelerating through stationary blades, decelerating through moving blades,
- ◆ no change in steam velocity across the stage, but
 - pressure decrease
 - temperature decrease
- ◆ This reflects the work performed in driving the rotor



Impulse principle

- ◆ Impulse turbine (dt. Gleichdruckturbine)
- ◆ Steam expansion takes place in stationary nozzles
- ◆ Fixed nozzles – increasing the speed of working medium
- ◆ Rotor blade are shaped like buckets
- ◆ Absorbing kinetic energy into mechanical work – results in shaft rotation
- ◆ Pressure drop across stationary blades together with an velocity increase



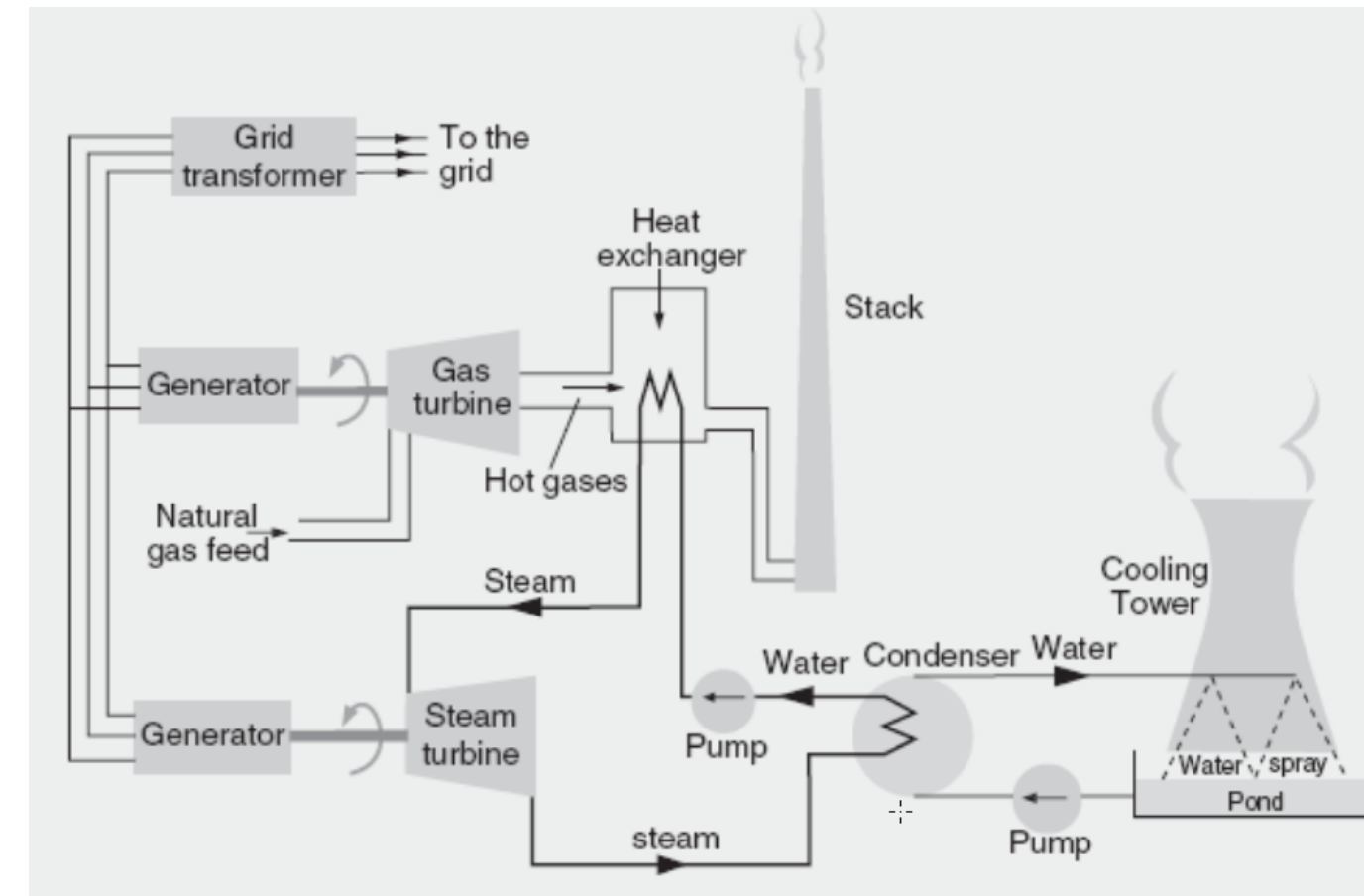
Impulse / Reaction Principle

- ◆ Impulse design
 - high pressure region
 - small volume of steam
- ◆ Reaction design
 - low pressure region
 - large volume of steam

- ◆ Most modern turbines are a combination

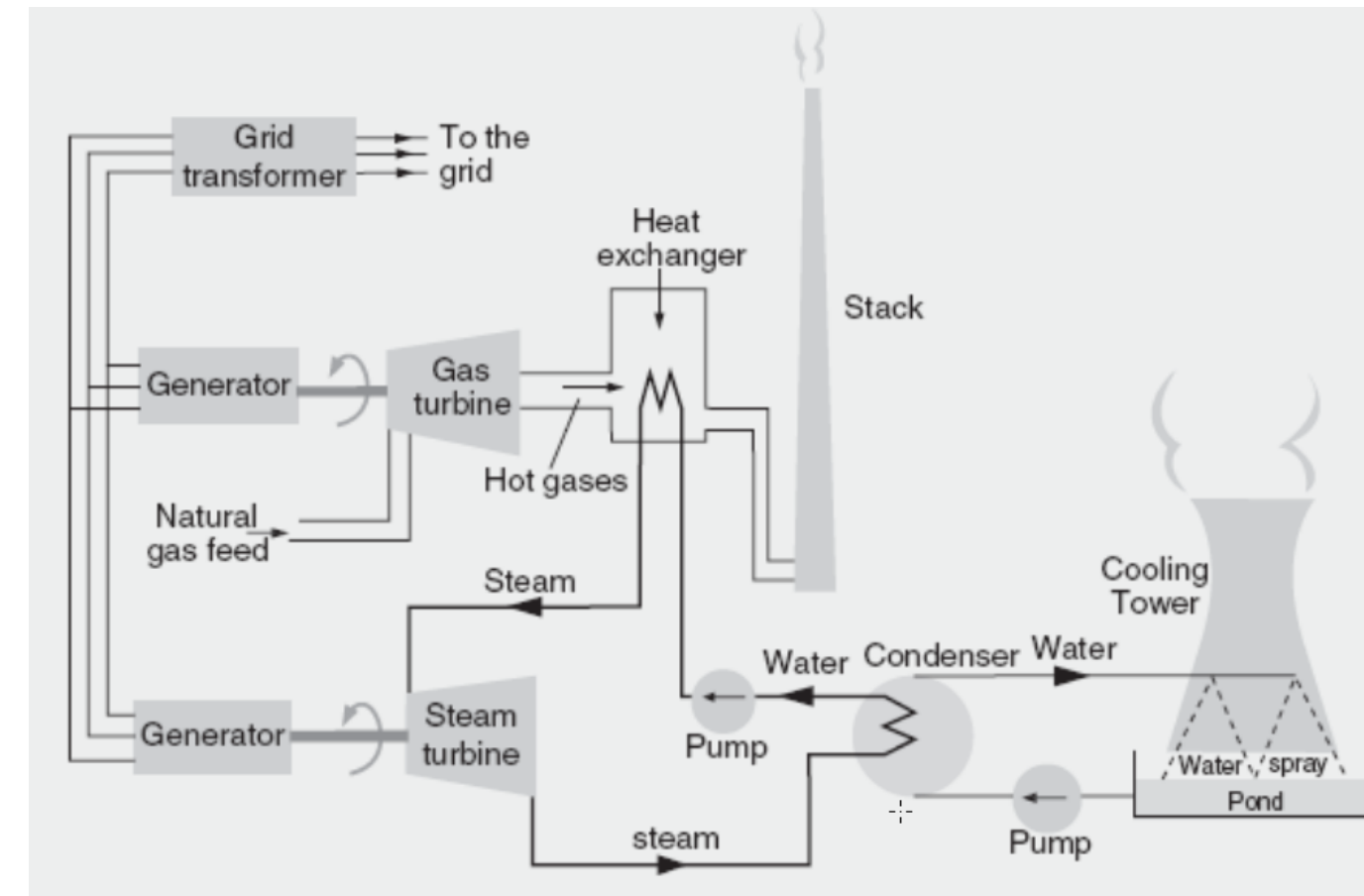
Combined cycle gas turbine CCGT

- ♦ Low price for natural gas
 - gas fired power plants increasingly used
- ♦ Gas turbines
 - higher efficiency than coal power plants
 - still sufficient heat in the exhaust to heat a boiler
- ♦ Boiler is used to drive a steam turbine
- ♦ Efficiency ~ 60%



Combined cycle gas turbine CCGT

- ◆ Combined cycle gas turbine feature
 - High efficiency possible with large units
 - Smaller units can start up quickly
 - Ability to run on light oil (if gas supply is interrupted)
 - Fast installation (modular units, factory supplied units)
 - Flexible possibility for load following
- ◆ Start up time
 - small gas turbine 2-3 min
 - steam turbine 20 min



Nuclear Power Plants

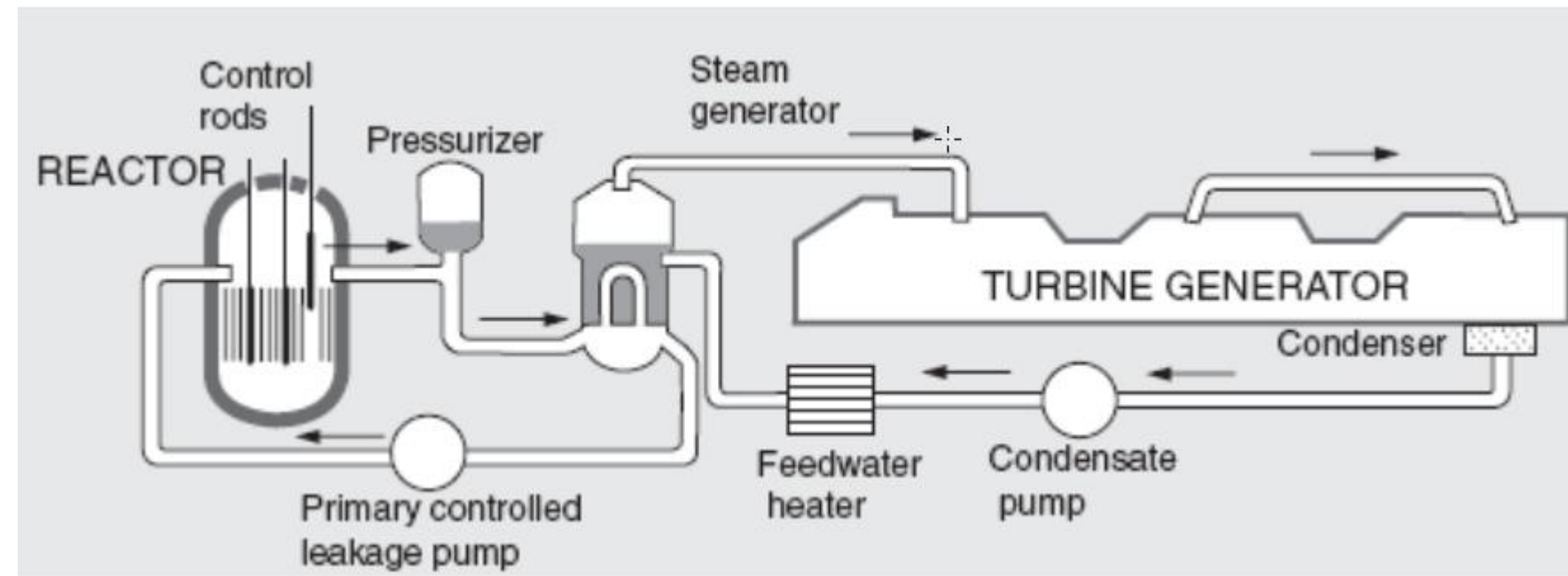
- ♦ Heat comes from fission reaction (dt. Kernspaltung)
- ♦ Splitting uranium atoms
- ♦ Uranium metal consists of
 - Isotope ^{238}U (99,3%)
 - Isotope ^{235}U (0,7%)
- ♦ ^{235}U is fissile (dt. spaltbar)
- ♦ ^{235}U gets stuck by slow-moving neutrons
 - splits into two substantial fragments
 - plus several neutrons
 - 3×10^{-11} Joule of kinetic energy
- ♦ Fast moving fragments hit surrounding atoms producing heat
- ♦ Number of neutrons increases, causing chain reaction

Nuclear Power Plants

- ♦ Basic reactor consists of fuel in form of rods or pallets
- ♦ Which are situated in an environment (moderator) which will
 - slow down the neutrons and fission products
 - heat is evolved
- ♦ Moderator can be light or heavy water or graphite
- ♦ Light water – conventional H_2O
- ♦ Heavy water – Deuterium Oxide (D_2O)
- ♦ Also moveable rods are in a moderator
 - absorb neutrons
 - control over the fission process

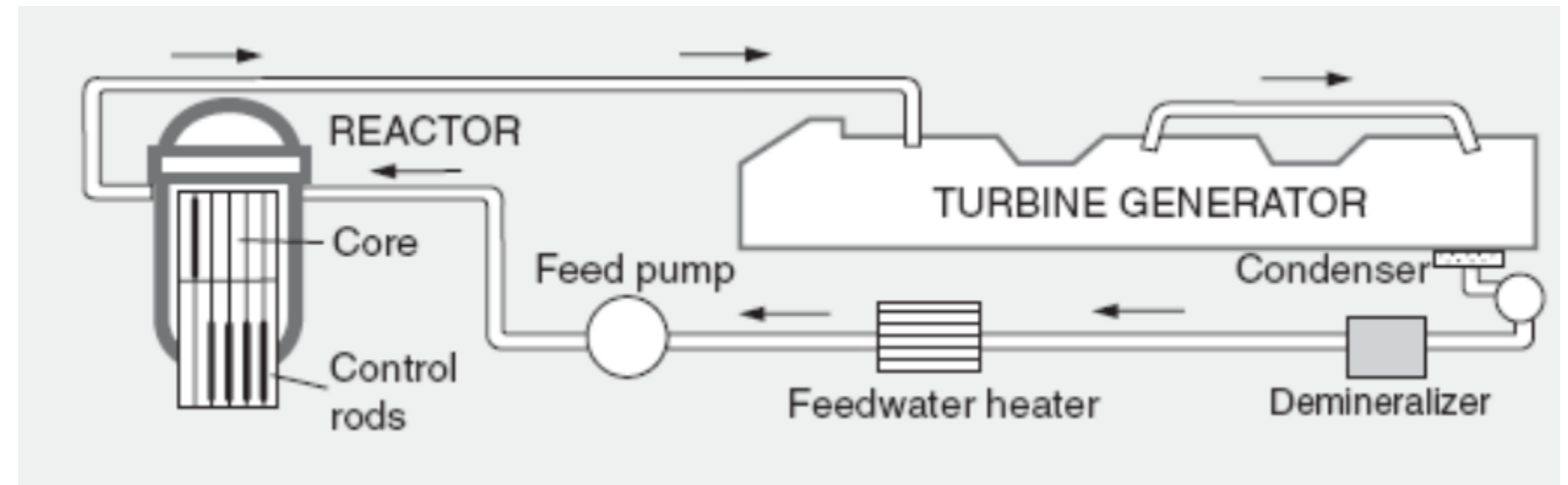
Pressurized Water Reactor (PWR)

- ◆ Pressurized water is pumped through the reactor and acts as coolant and moderator
- ◆ Water heated up to 315 °C and 150 bar pressure
- ◆ Leaves boiler to heat exchanger
- ◆ Second hydraulic circuit feeds steam to the turbine
- ◆ Fuel is in form of pellets of uranium dioxide in bundles of Zirconium alloy



Boiling water reactor (BWR)

- ◆ Developed later than the pressurized water type
- ◆ Inside the reactor, heat is transferred to boiling water at a pressure of 75 bar



Thermal power plant control

- ◆ Two main tasks of a thermal power plant controller
 - Boiler control
 - Turbine generator control

- ◆ Control strategies
 - Boiler following
 - Turbine following
 - Simultaneously

Boiler following

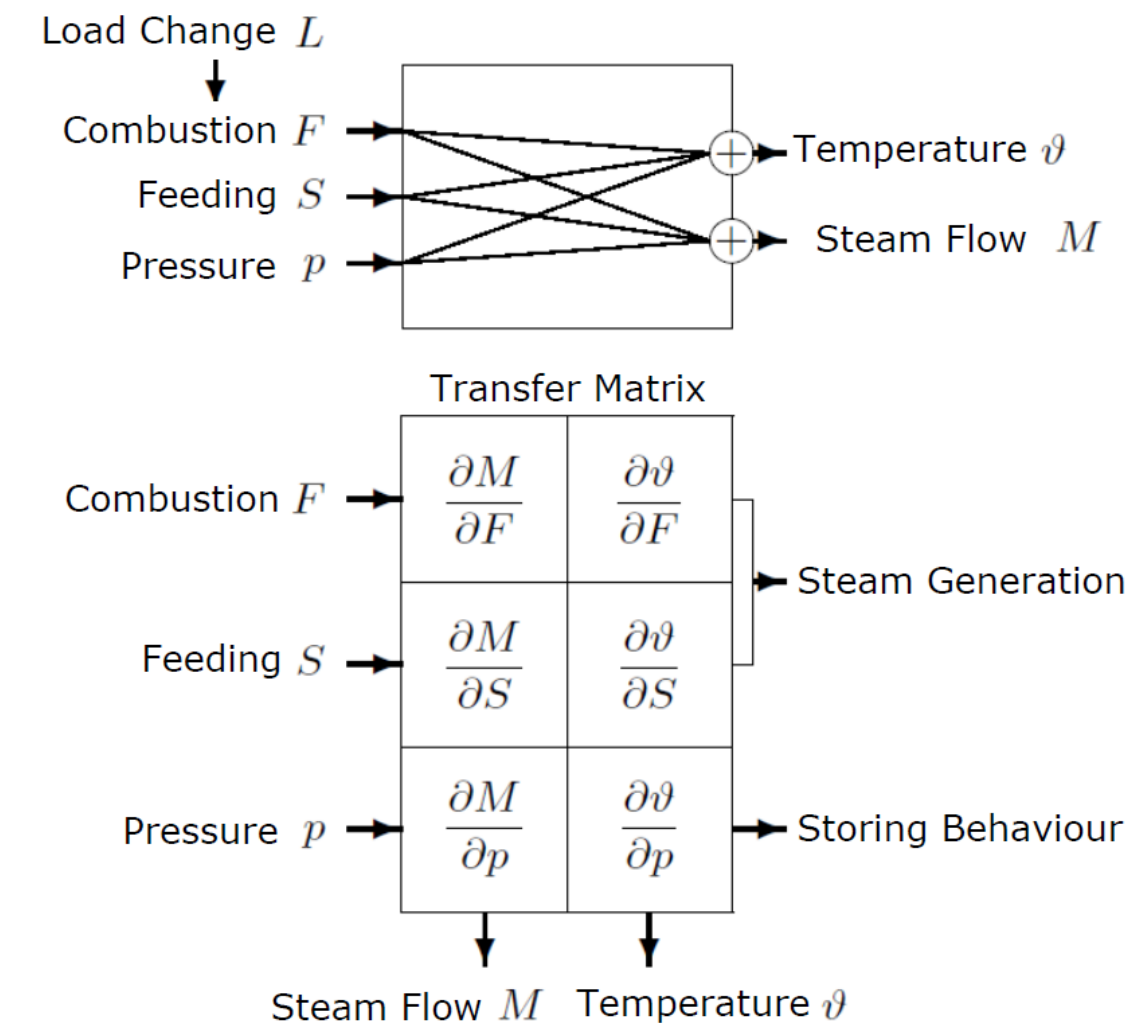
- ♦ In case of load change
 - Turbine governing valves opened
 - Causes a change in steam pressure
 - Triggers the boiler controller to adjust firing and steam sub process

Turbine following / Simultaneously

- ♦ Turbine controller follows boiler control
- ♦ In case of a load change
 - boiler controller will adapt firing and steam subsystem which results in a change of steam pressure
 - This triggers the turbine control to adapt the throttle pressure
- ♦ Simultaneously
 - Load change signals are simultaneously transmitted to
 - boiler control
 - turbine control
 - Each subsystem adjust itself within the defined limits

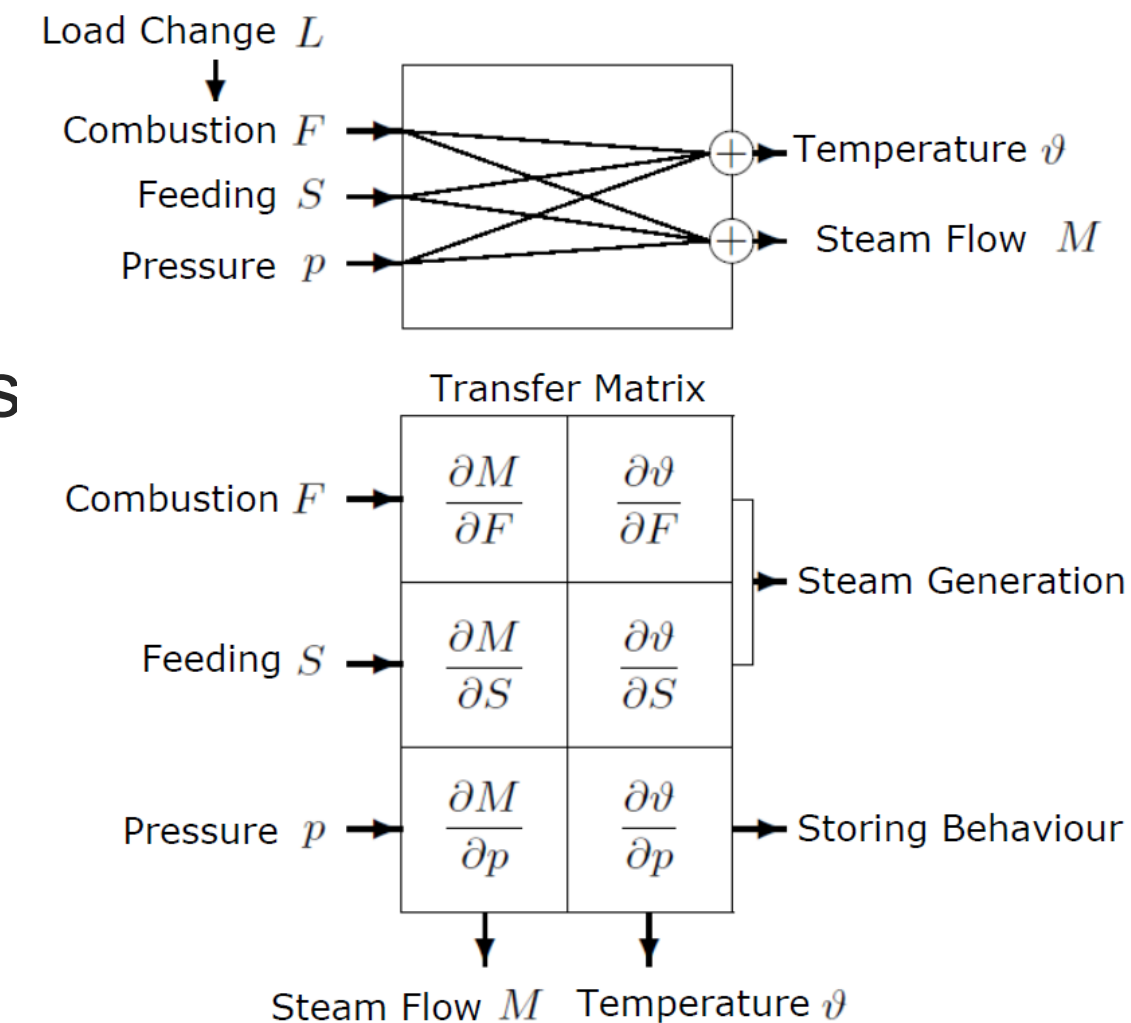
Boiler control

- ◆ Steam generator (aka boiler) consists of
 - firing system
 - steam generation
- ◆ Boiler controller
 - provide the required steam flow to the turbine
 - Three input values
 - load change
 - feeding
 - pressure
 - two output values
 - steam flow
 - temperature



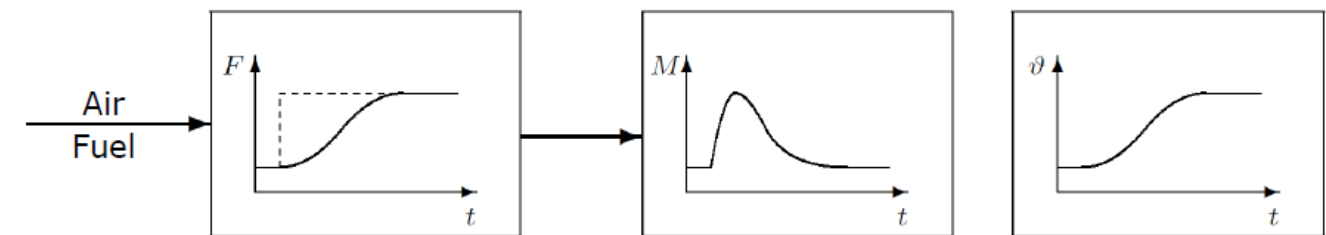
Boiler control

- ◆ Considered linearized relations
 - six transfer functions
 - usually grouped as matrix
- ◆ Defining the pressure as the input value for this process
 - Steam generation based
 - combustion
 - water feed-in
 - Storing behavior
 - pressure change at constant combustion
 - water feeding



Describe transfer function

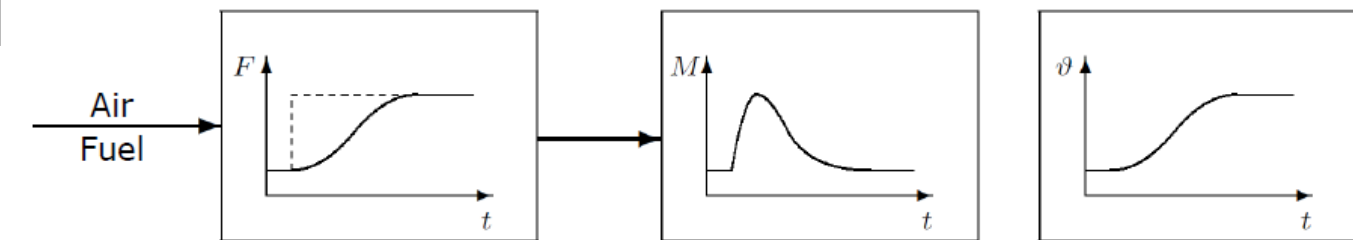
- ♦ Step response used to describe transfer function
- ♦ Steam generation step change
 - combustion increased by increasing fuel and air
 - Causes expansion of working medium
 - steam exhaust at the top of the boiler
 - Change finished as soon as the new temperature balance is reached
- ♦ Delay in temperature increase because flow of new working medium neutralizes the effect of increased heating for a certain time



Describe transfer function

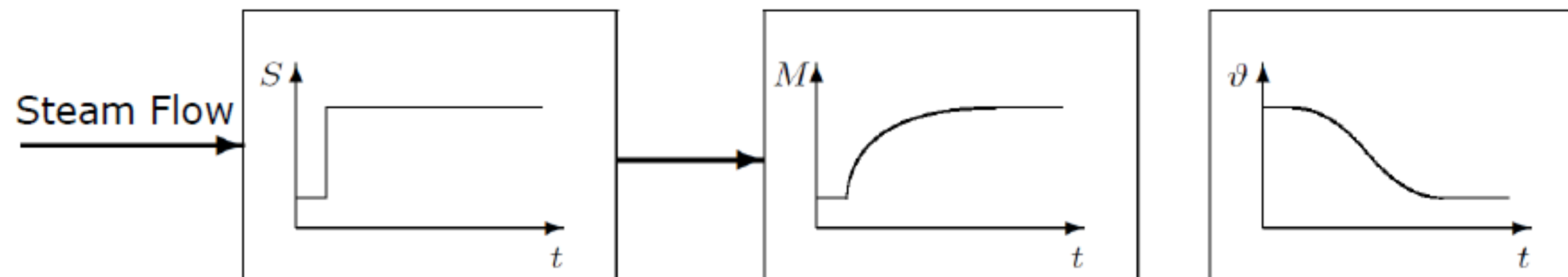
♦ Thus

- Suitable to control **steam flow** for a **short period** of time but not for steady state
- Suitable to control steam **temperature** for steady state but **not** for a **short period of time**



Feed-in change

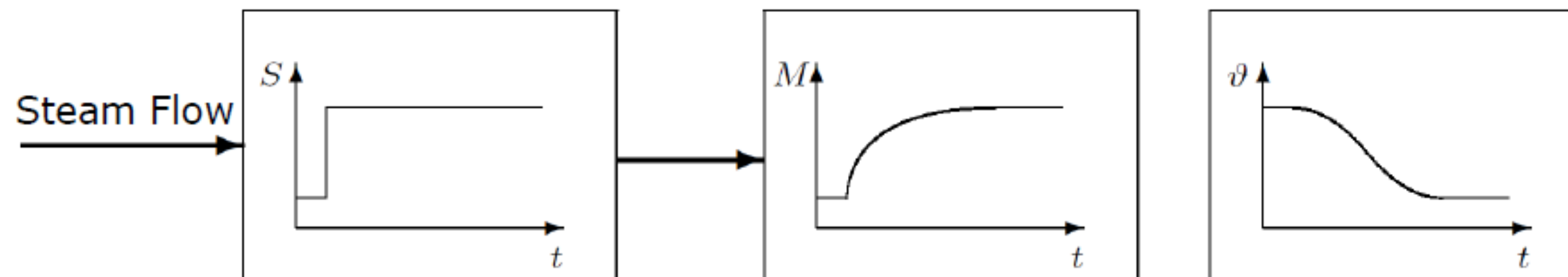
- ◆ Increasing the feeding
 - pressure decreases because of increased flow velocity along the boiler
 - Cooling effect
 - Leads to higher working medium mass which
 - effects steam flow
 - temperature



Feed-in change

◆ Thus

- Steam flow and temperature are adapted in steady state
- Because of temperature delay it is not suitable to use it for short term adaptations



Boiler control

- ♦ Control target
 - Minimize temperature change
 - Fast steam generation
- ♦ Temperature response identical for feed-in and steam generation
 - *The feed-in water flow has to be changed together with the combustion power change*
- ♦ Step response of feed-in change and combustion power change are compensation each other
 - *In case of a load change the feed-in flow in the combustion has to follow chronologically*

Turbine control

- ♦ Power output of the turbine is controlled by varying the flow rate of the steam input

$$P_T = \eta_T \dot{m} \Delta h$$

P_T	Turbine power
η_T	Turbine efficiency
$\dot{m}$	Flow rate
Δh	Enthalpy change

Turbine control

- ♦ Throttle valves are used to control the flow rate m by manipulating the cross-section of the turbine inlet valves.
- ♦ To operate the turbine in partial load condition, the throttle valves are reducing the pressure
- ♦ This causes
 - pressure change
 - temperature change

Turbine control

- ♦ Temperature change causes tension in the thick walled components
- ♦ Efficiency is reduced because the enthalpy change within the turbine is reduced as well
- ♦ Alternatively to throttle valve control
 - Only partially use the turbine
 - Certain groups of nozzles are activated / de-activated



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Andreas Nenning
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Course Number 074742056201